

GLOBAL HÖLDER ESTIMATES FOR LINEARIZED MONGE–AMPÈRE EQUATIONS IN DIVERGENCE FORM WITH APPLICATIONS TO DUAL SEMIGEOSTROPHIC EQUATIONS AND PERIODIC HOMOGENIZATION

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ABSTRACT. We establish interior and global Hölder estimates for linearized Monge–Ampère equations in divergence form in all dimensions $n \geq 3$, when the Hessian determinant of the convex Monge–Ampère potential is bounded above and below by positive constants and the vector field on the right-hand side is bounded. The estimates use the L^p norm of the solution for any $p > 1$. A key ingredient is an L^1 estimate, with appropriate decay rates, for the gradient of the Green’s function of the linearized Monge–Ampère operator in small sections, uniform with respect to the pole. As applications, we obtain uniform Hölder estimates for the time derivatives of the primal and dual potentials in the three-dimensional periodic dual semigeostrophic system when the initial density is bounded away from zero and infinity, and establish a linear convergence rate for periodic homogenization of the Monge–Ampère equation.

1. INTRODUCTION AND STATEMENT OF THE MAIN RESULTS

In this paper, we establish interior and global Hölder estimates for linearized Monge–Ampère equations in divergence form in all dimensions $n \geq 3$ where the convex Monge–Ampère potential has Hessian determinant bounded between two positive constants and the vector field on the right-hand side is bounded. We apply our results to obtain uniform Hölder estimates for the time derivatives of the primal and dual potentials in the three-dimensional periodic dual semigeostrophic system when the initial density is bounded away from zero and infinity, and to establish a linear convergence rate in periodic homogenization for the Monge–Ampère equation in all dimensions.

Let $\Omega \subset \mathbb{R}^n$ be a convex domain, and let $u \in C^3(\Omega)$ be convex with

$$(1.1) \quad 0 < \lambda \leq \det D^2 u \leq \Lambda \quad \text{in } \Omega.$$

Throughout, the cofactor matrix of the Hessian matrix $D^2 u = (D_{ij} u)_{1 \leq i, j \leq n}$ is denoted by

$$U = (U^{ij})_{1 \leq i, j \leq n} = \text{cof } D^2 u = (\det D^2 u)(D^2 u)^{-1}.$$

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It is positive definite and divergence-free; see [26, Lemma 2.56]. The cofactor matrix comes from linearizing the Monge–Ampère operator:

$$U = \frac{\partial \det D^2 u}{\partial (D^2 u)}.$$

We are interested in obtaining interior and global Hölder estimates for the linearized Monge–Ampère equations in divergence form

$$(1.2) \quad L_u v = \operatorname{div} \mathbf{F}$$

with a bounded vector field $\mathbf{F}$, where L_u is the linearized Monge–Ampère operator associated with the convex Monge–Ampère potential u :

$$L_u v := \operatorname{div}(U D v) = U^{ij} D_{ij} v.$$

This type of equation appears in singular Abreu’s equation [19, 24, 25, 28] and semigeotrophic equations in meteorology [1–3, 7–10, 22, 30, 32] to be described in the next subsection. For more information on the linearized Monge–Ampère equations, we refer to [26] and [16] and the references therein.

1.1. Hölder estimates for linearized Monge–Ampère equations in divergence form.

Caffarelli and Gutiérrez [4] proved a fundamental interior Harnack inequality for the homogeneous linearized Monge–Ampère equation (1.2) with $\mathbf{F} = 0$ under (1.1). Their result is an affine invariant version of the classical Harnack inequality of Moser, and Krylov and Safonov for linear, uniformly elliptic equations in divergence form and nondivergence form, respectively. Due to the affine invariance of the linearized Monge–Ampère equations, Euclidean balls were replaced by sections of the Monge–Ampère potential function in [4]. Sections are sublevel sets of convex functions after subtracting their supporting hyperplanes.

For (1.2) with a bounded vector field $\mathbf{F}$, Le [22] established an interior Hölder estimate in dimension two. The higher-dimensional result when $n \geq 3$ in [26, Theorem 15.6] assumes an additional integrability condition on the Hessian $D^2 u$; Le asked in [26, Remark 15.10] if this additional condition can be removed. Wang [41, Theorem 1.5] obtained an interior Hölder estimate for (1.2) under $(D^2 u)^{1/2} \mathbf{F} \in L^q$, $q > n$, and asked in [41, Remark 1.6(2)] whether boundedness of $\mathbf{F}$ alone suffices. Related estimates for linearized Monge–Ampère equations with drifts were obtained by Kim [18].

Cui, Wang, and Zhou [6] developed interior Hölder estimates for $L_u v = \mu$ with measures μ having controlled growth on sections. Their application to bounded vector fields requires the divergence to have one sign; see their Theorem 1.4 and Remark 1.5. Gu and Le [16] used estimates for the Monge–Ampère gradient $(D^2 u)^{-1/2} D_x g$ of the Green’s function of the linearized Monge–Ampère operator L_u to give a unified, different proof of the interior Hölder estimates in [6, 41]. They also obtained a global version. Here, we treat arbitrary bounded vector fields in all dimensions $n \geq 3$.

Our first result is concern with interior Hölder estimates.

Theorem 1.1 (Interior Hölder estimate with bounded vector fields). *Let $\Omega \subset \mathbb{R}^n$ ($n \geq 3$) be a convex domain, and let $u \in C^3(\Omega)$ be a convex function satisfying*

$$0 < \lambda \leq \det D^2 u \leq \Lambda < \infty.$$

Fix $x_0 \in \Omega$ and $h_0 > 0$ such that

$$(1.3) \quad S_{4h_0} \Subset \Omega, \quad \text{where } S_t := S_u(x_0, t) = \{x \in \bar{\Omega} : u(x) < u(x_0) + Du(x_0) \cdot (x - x_0) + t\}.$$

Let $p \in (1, \infty)$, $\mathbf{F} \in L^\infty(S_{2h_0}; \mathbb{R}^n)$, and $v \in W_{\text{loc}}^{1,2}(S_{2h_0}) \cap L^p(S_{2h_0})$ satisfy (1.2) weakly, that is,

$$\int_{S_{2h_0}} U Dv \cdot D\varphi \, dx = \int_{S_{2h_0}} \mathbf{F} \cdot D\varphi \, dx \quad \text{for all } \varphi \in C_c^\infty(S_{2h_0}).$$

There exist $\gamma = \gamma(n, \lambda, \Lambda) \in (0, 1)$ and $C = C(n, p, \lambda, \Lambda, h_0, \text{diam}(S_{4h_0})) > 0$ such that v has a representative with

$$(1.4) \quad \|v\|_{L^\infty(S_{h_0})} + [v]_{C^\gamma(S_{h_0})} \leq C \left(\|\mathbf{F}\|_{L^\infty(S_{2h_0})} + \|v\|_{L^p(S_{2h_0})} \right).$$

Theorem 1.1 answers the question in [26, Remark 15.10] and [41, Remark 1.6(2)] and removes the extra Hessian-integrability condition from the higher-dimensional part of [26, Theorem 15.6]. As stated, the constant in (1.4) does not depend on the ellipticity ratio of the cofactor matrix U on an interior compact set.

In Theorem 4.3, we extend the interior Hölder estimates in Theorem 1.1 to the case of L^q vector fields where q is large enough, depending only on n, λ , and Λ .

An important ingredient in the proof of Theorem 1.1 is the L^1 estimate, with appropriate decays, for the gradient of the Green's function of the linearized Monge–Ampère operator on small sections. For previous studies on the Green's function, we refer the readers to [16, 20–23, 26, 33–35, 38].

Proposition 1.2 (L^1 estimate for the gradient of the Green's function). *Assume $n \geq 3$, (1.1) and (1.3). Let $S = S_u(z, h)$ satisfy $S_u(z, 2h) \Subset S_{3h_0/2}$. Let $g_S(\cdot, y)$ be the Green's function of the linearized Monge–Ampère operator $L_u = D_i(U^{ij}D_j)$ in V with pole $y \in S$, that is,*

$$-D_i(U^{ij}D_j g_S(\cdot, y)) = \delta_y \quad \text{in } S, \quad g_S(\cdot, y) = 0 \quad \text{on } \partial S.$$

Then, there is a constant $\beta = \beta(n, \lambda, \Lambda) \in (0, 1)$ such that

$$(1.5) \quad \sup_{y \in S} \int_S |D_x g_S(x, y)| \, dx \leq Ch^\beta, \quad C = C(n, \lambda, \Lambda, h_0, \text{diam}(S_{4h_0})).$$

Remark 1.3. The estimate (1.5) is uniform even when the pole approaches ∂S . Its proof uses the geometry of sections of the Monge–Ampère equation [26, Chapter 5] and the Green's function estimates in [26, Chapter 14]. We give here some heuristic ideas. We estimate the Euclidean gradient Dg of the Green's function $g := g_S(\cdot, y)$ via the Monge–Ampère gradient $(D^2u)^{-1/2}Dg$ with a possibly large prefactor being the Laplacian of u (see (2.8)):

$$|Dg|^2 \leq \lambda^{-1}(\Delta u)U^{ij}D_i g D_j g \equiv \lambda^{-1}(\Delta u)(\det D^2u)|(D^2u)^{-1/2}Dg|^2.$$

The whole section S is foliated by sublevel sets $S_k = \{g \leq k\}$ of the Green's function. It is interesting to note that $U^{ij}D_i g D_j g$ gives total mass k on S_k , and this can be large. However, this is compensated by a localization property which says that the level-band $S_k \setminus S_{k/2}$ is contained in a section of u with height of order $k^{-\frac{2}{n-2}}$ (see (2.6)); furthermore, on this section, the Laplacian measure $\Delta u \, dx$ can be shown to have total measure of order $k^{-\frac{2}{n-2}(\frac{n-2}{2} + \frac{2\alpha}{1+\alpha})}$ (see (3.1)) where $\alpha = \alpha(n, \lambda, \Lambda) \in (0, 1)$ is the exponent in the $C^{1,\alpha}$ estimates for u . Via the Hölder inequality, this indicates that the total mass of $|Dg|$ in $S_k \setminus S_{k/2}$ decays when k is large. A dyadic summation over level-bands gives the desired result.

We also obtain global results under suitable assumptions on the domain and boundary data. In particular, a global extension of Theorem 1.1 is the following.

Theorem 1.4 (Global Hölder estimate with L^q vector fields). *Let Ω be a uniformly convex domain in $\mathbb{R}^n$ ($n \geq 3$), that is, for all $z \in \partial\Omega$, there is a ball $B_R(z_0)$ such that $\Omega \subset B_R(z_0)$ and $\partial\Omega \cap \partial B_R(z_0) = \{z\}$ for some uniform convexity radius $R > 0$. Let $\partial\Omega \in C^3$ and $u \in C^{1,1}(\overline{\Omega}) \cap C^3(\Omega)$ be a convex function satisfying*

$$\lambda \leq \det D^2 u \leq \Lambda \quad \text{and} \quad u|_{\partial\Omega} \in C^3.$$

Let $q > q_b = q_b(n, \lambda, \Lambda) > n$, $\sigma \in (0, 1]$, $\mathbf{F} \in L^q(\Omega; \mathbb{R}^n)$, and $v \in W^{1,2}(\Omega) \cap C(\overline{\Omega})$ satisfy

$$L_u v = \operatorname{div} \mathbf{F} \quad \text{weakly in } \Omega, \quad v = \phi \quad \text{on } \partial\Omega,$$

where $\phi \in C^\sigma(\partial\Omega)$. Then there exist $\gamma \in (0, 1)$ and $C > 0$, which depend only on $n, \lambda, \Lambda, q, \sigma, \|u\|_{C^3(\partial\Omega)}$, R , and the C^3 regularity of $\partial\Omega$, such that

$$(1.6) \quad \|v\|_{C^\gamma(\overline{\Omega})} \leq C \left(\|\phi\|_{C^\sigma(\partial\Omega)} + \|\mathbf{F}\|_{L^q(\Omega)} \right).$$

Remark 1.5.

- (i) (Two dimensions) We state our results including Theorems 1.1 and 1.4 for dimensions $n \geq 3$. However, the methods of the proofs are applicable to the two-dimensional case. For this, we only need to change the localization of the superlevel sets of the Green's function of the Monge–Ampère operator. In particular, when $n = 2$, we need to replace (2.6) by

$$\{x \in S : g_S(x, y) > k\} \subset S_u(y, 2\eta h 2^{-k/\tau}),$$

and (5.5) by

$$\{x \in V : g_V(x, y) > k\} \subset S_u(y, h 2^{-k/\bar{C}}),$$

if V is contained in a section of u with height h .

- (ii) (Global assumptions) In the proof of Theorem 1.4, the refined global analysis of the Green's function plays an important role. This was developed in [26, Section 14.4] under the global assumption $u \in C^{1,1}(\overline{\Omega})$ on the convex potential function. However, our global Hölder estimates do not depend on the bounds on the Hessian of u . This point will be crucial in establishing the nonlinear stability result in Theorem 1.8 via approximations.

In the next two subsections, we will describe some applications of Theorems 1.1 and 1.4.

1.2. Application to dual semigeostrophic equations. Our first application is concerned with the semigeostrophic equations which we give a brief introduction here.

The semigeostrophic equations are simplified models used in meteorology to describe large-scale atmospheric flows, particularly in mid-latitude weather systems where the Earth's rotation plays a crucial role. As explained by Benamou and Brenier [3, Section 2.2], Loeper [31, Section 1.1], and Cullen [7], these equations can be formally derived from the three-dimensional incompressible Euler equations under the Boussinesq and hydrostatic approximations, in a regime where the Coriolis force is dominant. For large-scale atmospheric motions, the Coriolis effect dominates the nonlinear advection term, and much of the mathematical literature has focused on the two-dimensional periodic setting; see, for instance, [1, 22]. The three-dimensional case is physically relevant but involves additional difficulties; see [2]. Here we work on the three-dimensional torus $\mathbb{T}^3 := \mathbb{R}^3/\mathbb{Z}^3$.

On $\mathbb{T}^3$, the semigeostrophic equations take the form

$$(1.7) \quad \left\{ \begin{array}{l} \partial_t u_t^g + (u_t \cdot D) u_t^g + D p_t = -J u_t + m_t e_3, \\ \partial_t m_t + (u_t \cdot D) m_t = 0, \\ u_t^g = J D p_t, \\ \operatorname{div} u_t = 0, \\ p_{t=0} = p^0, \end{array} \right.$$

where $e_3 = (0, 0, 1)^T$ and

$$J = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The functions u_t , p_t , and m_t denote the velocity, pressure, and buoyancy, respectively, and u_t^g is the semigeostrophic wind. We use the notation $f_t = f(t, \cdot)$.

Substituting $u_t^g = J D p_t$ and introducing

$$P_t(x) := p_t(x) + \frac{1}{2}(x_1^2 + x_2^2),$$

the system (1.7) can be rewritten as

$$(1.7) \quad \left\{ \begin{array}{l} \partial_t D P_t + D^2 P_t u_t = J(D P_t - x), \\ \operatorname{div} u_t = 0, \\ P_{t=0}(x) = p^0(x) + \frac{1}{2}(x_1^2 + x_2^2). \end{array} \right.$$

Energetic considerations make convexity of P_t natural; see [7, Section 3.2]. Since the existence theory for the primal system is difficult, one passes to the dual formulation. If P_t^* denotes the Legendre transform of P_t , the dual semigeostrophic equations are

$$(1.8) \quad \left\{ \begin{array}{l} \partial_t \rho_t + \operatorname{div}(\rho_t U_t) = 0, \\ U_t(x) = J(x - D P_t^*(x)), \\ \det D^2 P_t^* = \rho_t, \\ \rho_{t=0} = \rho^0, \end{array} \right. \quad (t, x) \in (0, \infty) \times \mathbb{T}^3,$$

with P_t^* convex and

$$(1.9) \quad P_t^*(x) - \frac{1}{2}|x|^2 \quad \text{is } \mathbb{Z}^3\text{-periodic.}$$

The Monge–Ampère equation in (1.8) is understood in the Aleksandrov sense for weak solutions. This dual formulation and its relation to the primal variables are discussed in [2, 3, 30]. Formally, once a sufficiently regular dual solution is available, one can recover the physical variables through the Legendre transform. The main obstruction is therefore to obtain enough regularity for the dual potential.

The spatial regularity is by now well understood through the regularity theory of the Monge–Ampère equation; see, for instance, [13, 14, 26]. It is then natural to investigate regularity in time. Differentiating $\det D^2 P_t^* = \rho_t$ with respect to t and using the continuity equation in (1.8), we obtain

$$\operatorname{cof}(D^2 P_t^*)^{ij} D_{ij}(\partial_t P_t^*) = \partial_t \rho_t = \operatorname{div}(-\rho_t U_t) =: \operatorname{div} F_t.$$

Since the cofactor matrix is divergence-free, $v_t := \partial_t P_t^*$ satisfies the linearized Monge–Ampère equation

$$(1.10) \quad \operatorname{div}(\rho_t(D^2 P_t^*)^{-1} D \partial_t P_t^*) = \operatorname{div} F_t, \quad F_t = -\rho_t U_t.$$

If the initial density ρ^0 is bounded from below and above by positive constants, we can show that F_t is uniformly bounded in $L^\infty(\mathbb{T}^3)$. Moreover, the time-regularity estimates developed in [1, 2, 30] give, after fixing the additive constant of P_t^* , an L^p bound for $\partial_t P_t^*$ for some $p > 1$. Thus the Hölder continuity of $\partial_t P_t^*$ reduces exactly to the estimate of Theorem 1.1.

Regarding previous time-regularity results, Loeper [30, Theorem 9.2] proved that if the initial density is sufficiently close to a constant, namely $1 - \varepsilon_0 \leq \rho^0 \leq 1 + \varepsilon_0$ on $\mathbb{T}^3$, then $\partial_t P_t$ and $\partial_t P_t^*$ are uniformly Hölder continuous in time with values in space. Le [22, Theorem 1.2] later removed the small-oscillation assumption on the two-dimensional torus. Combining Theorem 1.1 with the three-dimensional estimates in [2], and otherwise following the argument of [22, Section 4] (see also [26, Theorem 15.13]), gives the following three-dimensional result.

Theorem 1.6 (Hölder regularity of three-dimensional dual semigeostrophic equations). *Let ρ^0 be a probability density on $\mathbb{T}^3$ satisfying*

$$0 < \lambda \leq \rho^0 \leq \Lambda < \infty.$$

Let (ρ_t, P_t^) be a solution of (1.8)–(1.9), and let P_t be the Legendre transform of P_t^* . Then there exist $\alpha = \alpha(\lambda, \Lambda) \in (0, 1)$ and $C = C(\lambda, \Lambda) > 0$ such that*

$$(1.11) \quad \|\partial_t P_t^*\|_{L^\infty((0, \infty); C^\alpha(\mathbb{T}^3))} + \|\partial_t P_t\|_{L^\infty((0, \infty); C^\alpha(\mathbb{T}^3))} \leq C.$$

The proof is given in Section 6.1. The crucial point, compared with the earlier argument, is that Theorem 1.1 only requires an L^p norm of the solution for some $p > 1$. Hence, the known estimate $D \partial_t P_t^* \in L^{1+\kappa}$, together with Poincaré’s inequality after normalization, supplies exactly the integrability needed for (1.4).

1.3. Application to quantitative homogenization for the Monge–Ampère equation.

A second application concerns quantitative homogenization for the Monge–Ampère equation. Here the domain and the boundary data are fixed, while the prescribed Monge–Ampère density oscillates on a scale tending to zero. The question is how these oscillations affect the convex solution at a fixed spatial scale. Unlike a perturbation with small amplitude, a rapidly oscillating density need not be close to its average in any strong L^p norm. An effective stability estimate must therefore detect cancellation in the density, rather than only its pointwise size.

Let $\Omega \subset \mathbb{R}^n$ ($n \geq 2$) be a bounded smooth uniformly convex domain, and $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ so it is $[0, 1]^n$ with opposite faces identified. Let $f \in L^\infty(\mathbb{T}^n)$, with periodic extension to $\mathbb{R}^n$, satisfy

$$(1.12) \quad 0 < \lambda \leq f \leq \Lambda, \quad \bar{f} := \int_{[0, 1]^n} f(y) \, dy.$$

Fix also a boundary datum $\varphi \in C^3(\partial\Omega)$. We consider the convex Aleksandrov solutions of

$$(1.13) \quad \begin{cases} \det D^2 u_\varepsilon = f(x/\varepsilon) & \text{in } \Omega, \\ \det D^2 \bar{u} = \bar{f} & \text{in } \Omega, \\ u_\varepsilon = \bar{u} = \varphi & \text{on } \partial\Omega. \end{cases}$$

Periodic averaging (see [42, pp. 5–6]) gives $f(\cdot/\varepsilon) \rightharpoonup^* \bar{f}$ in $L^\infty(\Omega)$. Together with the compactness and weak continuity of Monge–Ampère measures, this identifies $\bar{u}$ as the uniform

limit of u_ε ; see [26, Chapter 3]. This qualitative argument does not provide a convergence rate.

Caffarelli and Li [5] established a quantitative estimate in their study of entire solutions with periodic Monge–Ampère density. For a normalized convex domain $B_1 \subset O \subset B_n$ and a positive periodic density g of cell average one, let w and $\bar{w}$ have zero boundary values on ∂O and Monge–Ampère densities g and 1, respectively. Their Theorem 0.3 gives

$$\|w - \bar{w}\|_{L^\infty(O)} \leq C \sum_{i=1}^n |\ell_i|^b,$$

where $\ell_1, \dots, \ell_n$ are linearly independent period vectors and $C, b > 0$ depend only on n and an upper bound for g . Their Remark 0.7 extends this estimate to bounded periodic densities by approximation. For periods εe_i , it gives a rate $C\varepsilon^b$. The proof uses periodic correctors and interior derivative estimates for the constant-density solution. Li and Lu [29] subsequently proved the existence of periodic correctors and extended the entire-solution classification to measurable Monge–Ampère densities bounded away from zero. Recently, Jin, Li, Tran and Tu [17] classified global convex solutions to $\det D^2u = \mu$ for all nonzero, nonnegative locally finite periodic Borel measures μ on $\mathbb{R}^n$. The above-mentioned results concern the structure of entire solutions. Here, we seek a uniform rate for the fixed-domain Dirichlet problem.

The difficulty is not the identification of the effective density. Indeed, for every $1 \leq p < \infty$, periodic averaging gives

$$\lim_{\varepsilon \rightarrow 0^+} \|f(\cdot/\varepsilon) - \bar{f}\|_{L^p(\Omega)}^p = |\Omega| \int_{[0,1]^n} |f(y) - \bar{f}|^p dy,$$

which is positive unless f is constant almost everywhere. Thus, an estimate depending only on this strong norm cannot capture the small period. Moreover, the Hessian determinant bounds alone do not control the ellipticity ratio of coefficient matrix of the linearized operator. Consequently, classical uniformly elliptic estimates do not directly give bounds with constants depending only on the Monge–Ampère density bounds.

We state an application to the periodic homogenization of the Monge–Ampère equation in all dimensions $n \geq 2$. For $n \geq 3$, the global Hölder estimate needed below is provided by Theorem 1.4; in dimension two, we use [23, Theorem 1.2] (see also [26, Theorem 15.9]). Along the density homotopy, the boundary datum is fixed, so its derivative has zero boundary trace. The density-homotopy argument gives the following global estimate.

Theorem 1.7 (Linear rate in periodic homogenization). *Let $\Omega \subset \mathbb{R}^n$ ($n \geq 2$) be a bounded smooth uniformly convex domain. Let $\varphi \in C^3(\partial\Omega)$, and let $f \in L^\infty(\mathbb{T}^n)$ satisfy (1.12). Let u_ε and $\bar{u}$ solve (1.13). There exist $\gamma \in (0, 1)$ and $C > 0$, depending only on $n, \lambda, \Lambda, \Omega$, and $\|\varphi\|_{C^3(\partial\Omega)}$, such that, for every $\varepsilon > 0$,*

$$(1.14) \quad \|u_\varepsilon - \bar{u}\|_{L^\infty(\Omega)} \leq C\varepsilon \|f - \bar{f}\|_{L^\infty(\mathbb{T}^n)}.$$

Moreover,

$$(1.15) \quad [u_\varepsilon - \bar{u}]_{C^\gamma(\bar{\Omega})} \leq C\varepsilon \|f - \bar{f}\|_{L^\infty(\mathbb{T}^n)}.$$

The above estimates give a linear rate in the uniformly nondegenerate class of Monge–Ampère equations on a fixed smooth uniformly convex domain. These assumptions are stronger than the domain assumptions in [5], whose normalized estimate also does not require a fixed positive lower density bound. We do not claim that the linear rate is optimal. Nevertheless,

this result is reminiscent of the recent work of Tran and Yu [39] on the periodic homogenization of convex Hamilton–Jacobi equations, where the optimal rate of convergence was shown to be linear. The global Hölder linear rate in (1.15) follows from the global nonlinear stability in Theorem 1.8; in particular, it is not obtained by interpolating (1.14) with a bound independent of ε , which would generally lose a power of ε .

In fact, the periodic homogenization follows from a global nonlinear stability estimate with negative-norm for the Monge–Ampère equation in Theorem 1.8 below that does not require periodicity.

For $f \in L^\infty(\Omega)$, set

$$\|f\|_{\text{div},\infty;\Omega} := \inf \left\{ \|\mathbf{F}\|_{L^\infty(\Omega)} : \mathbf{F} \in L^\infty(\Omega; \mathbb{R}^n), \quad \text{div } \mathbf{F} = f \text{ in } \mathcal{D}'(\Omega) \right\}.$$

In the set of competitors, no boundary condition is imposed on $\mathbf{F}$. The infimum is finite: extend f by zero to a box containing Ω and integrate in one coordinate to obtain a bounded vector field with divergence f . This simple argument also shows that

$$\|f\|_{\text{div},\infty;\Omega} \leq C(\Omega) \|f\|_{L^\infty(\Omega)}.$$

Theorem 1.8 (Nonlinear stability for the Monge–Ampère equation). *Let $\Omega \subset \mathbb{R}^n$ ($n \geq 2$) be a smooth uniformly convex domain. Let $0 < \lambda \leq \Lambda < \infty$ and $\varphi \in C^3(\partial\Omega)$. Let $f_0, f_1 \in L^\infty(\Omega)$ satisfy $\lambda \leq f_i \leq \Lambda$, and let $u_i \in C(\overline{\Omega})$ ($i = 0, 1$) be the convex Aleksandrov solution of*

$$\det D^2 u_i = f_i \quad \text{in } \Omega, \quad u_i = \varphi \quad \text{on } \partial\Omega.$$

Then, there exist $\gamma \in (0, 1)$ and $C > 0$, depending only on $n, \lambda, \Lambda, \Omega$, and $\|\varphi\|_{C^3(\partial\Omega)}$, such that

$$(1.16) \quad \|u_1 - u_0\|_{C^\gamma(\overline{\Omega})} \leq C \|f_1 - f_0\|_{\text{div},\infty;\Omega}.$$

Theorem 1.8 shows that the solution map $f \mapsto u$ is Lipschitz, with values in $C^\gamma(\overline{\Omega})$, with respect to a norm that measures the smallness of a bounded divergence primitive rather than the amplitude of the density difference. Equality of the total masses of f_0 and f_1 is not required.

For the periodic problem, the relevant smallness is elementary. The Poisson’s equation $\Delta_y \psi = f - \bar{f}$ on $\mathbb{T}^n$ has a zero-mean solution for which $G = D_y \psi$ satisfies

$$\text{div}_y G = f - \bar{f}, \quad \|G\|_{L^\infty(\mathbb{T}^n)} \leq C(n) \|f - \bar{f}\|_{L^\infty(\mathbb{T}^n)}.$$

The bound follows from the periodic $W^{2,p}$ estimate with $p > n$ and Sobolev embedding; see [15, Chapters 7 and 9]. Hence $\mathbf{F}_\varepsilon(x) := \varepsilon G(x/\varepsilon)$ satisfies

$$f(x/\varepsilon) - \bar{f} = \text{div } \mathbf{F}_\varepsilon, \quad \|\mathbf{F}_\varepsilon\|_{L^\infty(\Omega)} \leq C(n) \varepsilon \|f - \bar{f}\|_{L^\infty(\mathbb{T}^n)}.$$

This is where the small period enters the estimate.

To explain the nonlinear step, assume first that the densities f_0 and f_1 are smooth. Observe that $w := u_1 - u_0$ satisfies the linear elliptic equation

$$(1.17) \quad \left(\int_0^1 \text{cof}((1-s)D^2 u_0 + sD^2 u_1) ds \right)^{ij} D_{ij} w = f_1 - f_0.$$

This averaged coefficient matrix need not be the cofactor Hessian of a convex potential satisfying the same Hessian determinant bounds as those of u_0 and u_1 .

Instead, we interpolate the densities and solve

$$\det D^2 u_s = (1-s)f_0 + sf_1 \quad \text{in } \Omega, \quad u_s = \varphi \quad \text{on } \partial\Omega, \quad 0 \leq s \leq 1.$$

The global $C^{2,\alpha}(\overline{\Omega})$ solvability and the Implicit Function Theorem justify differentiation in s ; see [15, 40]. If $\operatorname{div} \mathbf{F} = f_1 - f_0$, then $v_s := \partial_s u_s$ solves

$$L_{u_s} v_s = \operatorname{div} \mathbf{F} \quad \text{in } \Omega, \quad v_s = 0 \quad \text{on } \partial\Omega.$$

The Hessian determinant bounds are preserved along this path. For $n \geq 3$, take $q = q_b + 1$ in Theorem 1.4. Since $\mathbf{F} \in L^\infty(\Omega)$, that theorem gives

$$\|v_s\|_{C^\gamma(\overline{\Omega})} \leq C \|\mathbf{F}\|_{L^q(\Omega)} \leq C \|\mathbf{F}\|_{L^\infty(\Omega)},$$

uniformly in s . In dimension two, the same global estimate follows from [23, Theorem 1.2]. Integrating $u_1 - u_0 = \int_0^1 v_s \, ds$ and minimizing over $\mathbf{F}$ give the global stability estimate.

Only the resulting nonlinear estimate is passed to bounded measurable densities, using smooth approximations that preserve the divergence identity. The proof, including this approximation and Theorem 1.7, is given in Section 6.2.

Remark 1.9. We make several remarks on Theorem 1.8.

- (i) (Negative norm) We emphasize that the main point in the nonlinear stability estimates in Theorem 1.8 is the *negative norm* $\|f_1 - f_0\|_{\operatorname{div}, \infty; \Omega}$ on the right-hand side. If this is replaced by the larger quantity $\|f_1 - f_0\|_{L^\infty(\Omega)}$, then the resulting estimate follows from the homotopy argument above and replacing Theorem 1.4 by the global Hölder estimates for the linearized Monge–Ampère equation with L^n right hand side; see [26, Theorem 13.2].
- (ii) ($C^{1,\alpha}$ stability with a small power rate) Under the hypotheses of Theorem 1.8, the functions u_0, u_1 are $C^{1,\beta}(\overline{\Omega})$ for some $\beta = \beta(n, \lambda, \Lambda) \in (0, 1)$; see [26, Theorem 9.5]. Let $\alpha \in (0, \beta)$. Then, using the interpolation inequality

$$\|u_1 - u_0\|_{C^{1,\alpha}(\overline{\Omega})} \leq C(n, \Omega, \alpha, \beta) \|u_1 - u_0\|_{L^\infty(\Omega)}^{\frac{\beta-\alpha}{1+\beta}} \|u_1 - u_0\|_{C^{1,\beta}(\overline{\Omega})}^{\frac{1+\alpha}{1+\beta}},$$

and the discussion in (i), we obtain the following $C^{1,\alpha}$ stability with a small power rate:

$$\|u_1 - u_0\|_{C^{1,\alpha}(\overline{\Omega})} \leq C(n, \Omega, \alpha, \beta, \lambda, \Lambda, \|\varphi\|_{C^3(\partial\Omega)}) \|f_1 - f_0\|_{L^\infty(\Omega)}^{\frac{\beta-\alpha}{1+\beta}}.$$

- (iii) ($C^{2,\beta}$ stability with a linear rate) If we impose additionally $f_0, f_1 \in C^\beta(\overline{\Omega})$, then, in view of [40, Theorem 1.1], $u_0, u_1 \in C^{2,\beta}(\overline{\Omega})$. Therefore, the averaged coefficient matrix in (1.17) is uniformly elliptic and belongs to $C^\beta(\overline{\Omega})$. Consequently, by the classical Schauder theory, instead of (1.16), we have a stronger estimate

$$\|u_1 - u_0\|_{C^{2,\beta}(\overline{\Omega})} \leq C \|f_1 - f_0\|_{C^\beta(\overline{\Omega})}.$$

- (iv) (Sobolev space stability) For the stability in Sobolev space $W^{2,\gamma}$ where $\gamma > 1$, see [11, Theorem 1.1] and [27, Proposition 3.18].

AI assistance. The authors have worked on linearized Monge–Ampère equations in divergence form during the last decade [6, 16, 19, 22, 23, 26]. The unconditional Hölder estimates previously obtained in this line of work were restricted to dimension two. A main obstacle to extending them to dimensions $n \geq 3$ was the lack of a suitable integrability estimate for the gradient of the Green’s function of the linearized Monge–Ampère operator. The argument leading to Proposition 1.2, which provides the missing L^1 estimate for the gradient of the Green’s function, was first produced in repeated interactions with OpenAI’s GPT-6 Astra on September 16, 2026. The authors subsequently checked the argument of Proposition 1.2 in detail, verified the proof independently, and take full responsibility for its correctness. This estimate supplied the key missing ingredient for completing the higher-dimensional argument. With it, the authors could complete our research program without further assistance.

The rest of the paper is organized as follows. We recall properties of sections of solutions to the Monge–Ampère equation and the Green’s functions of the linearized Monge–Ampère operator in Section 2. Section 3 is devoted to the proof of Proposition 1.2 and the uniform estimate for zero-boundary solutions. The proof of Theorem 1.1 is completed in Section 4 by a splitting argument. We then apply the resulting estimate to the three-dimensional periodic dual semigeostrophic equations in Section 6.1. The global higher integrability estimate for the gradient of the Green’s function and Theorem 1.4 are proved in Section 5, while Section 6.2 concludes with the negative-norm stability estimate and the periodic homogenization rate.

2. PRELIMINARIES

In this section, we recall some definitions and properties of sections of solutions to the Monge–Ampère equation and the Green’s functions of the linearized Monge–Ampère operator.

We recall the notion of sections of a convex function.

Definition 2.1 (Section). *Let Ω be a bounded convex domain in $\mathbb{R}^n$ ($n \geq 2$). Let $u \in C^1(\Omega) \cap C(\bar{\Omega})$ be a convex function and $h > 0$. If $x_0 \in \bar{\Omega}$, then the section of u centered at x_0 with height h is defined by*

$$S_u(x_0, h) := \{x \in \bar{\Omega} : u(x) < u(x_0) + Du(x_0) \cdot (x - x_0) + h\}.$$

In the case of $x_0 \in \partial\Omega$, we require that u is differentiable at x_0 .

Following [16, Definition 1.3], we define the notation of proper sets as follows.

Definition 2.2 (Proper set). *A nonempty open set $V \subset \mathbb{R}^n$ is called proper if it satisfies an exterior cone condition at every boundary point. Examples include intersections of a bounded convex domain $\Omega \subset \mathbb{R}^n$ with sections of a convex function $u \in C^1(\bar{\Omega})$ or open balls.*

The Green’s function of the linearized Monge–Ampère operator is defined as follows; see [26, Chapter 14].

Definition 2.3 (Green’s function of the linearized Monge–Ampère operator). *Let Ω be a bounded convex domain in $\mathbb{R}^n$ and $u \in C^3(\Omega)$ be a convex function satisfying (1.1). Let $V \subset \Omega$ be open. Let δ_y be the Dirac measure giving the unit mass to y . Then, for each $y \in V$, there exists a unique function $g_V(\cdot, y) : V \rightarrow [0, \infty]$ with the following properties:*

- (i) $g_V(\cdot, y) \in W_0^{1,q}(V) \cap W^{1,2}(V \setminus B_r(y))$ for all $q < \frac{n}{n-1}$ and all $r > 0$.
- (ii) $g_V(\cdot, y)$ is a weak solution of

$$-D_i \left(U^{ij} D_j g_V(\cdot, y) \right) = \delta_y \quad \text{in } V, \quad g_V(\cdot, y) = 0 \quad \text{on } \partial V,$$

that is, denoting $D_j g_V(x, y) = \frac{\partial}{\partial x_j} g_V(x, y)$, we have

$$(2.1) \quad \int_V U^{ij} D_j g_V(x, y) D_i \psi(x) dx = \psi(y) \quad \text{for all } \psi \in C_c^\infty(V).$$

We call $g_V(\cdot, y)$ the Green's function of the linearized Monge-Ampère operator $L_u = D_i(U^{ij} D_j)$ in V with pole y , and set $g_V(y, y) = +\infty$.

Remark 2.4 (Representation formula using the Green's function). Assume that $V \subset \mathbb{R}^n$ is proper. If $\varphi \in L^n(V)$, then there exists a unique solution $\psi \in W_{\text{loc}}^{2,n}(V) \cap W_0^{1,2}(V) \cap C(\bar{V})$ to

$$-U^{ij} D_{ij} \psi = \varphi \quad \text{in } V \quad \text{and } \psi = 0 \quad \text{on } \partial V.$$

Use ψ as a test function to (2.1), then the following holds (see also [26, Equation (14.5)]):

$$(2.2) \quad \psi(y) = \int_V g_V(\cdot, y) \varphi dx.$$

Remark 2.5. (Volume and interior $C^{1,\alpha}$ estimates) Let $u \in C(\Omega)$ be a strictly convex function satisfying (1.1) in the sense of Aleksandrov, where $\Omega \subset \mathbb{R}^n$ ($n \geq 3$). Then $u \in C^{1,\alpha}(\Omega)$ where $\alpha = \alpha(n, \lambda, \Lambda) \in (0, 1)$; see [26, Theorem 5.21]. For later references, we set

$$(2.3) \quad a = \frac{n-2}{2}, \quad \beta = \frac{\alpha}{1+\alpha}.$$

We use the following volume estimates (see [26, Lemma 5.6 (i)])

$$(2.4) \quad C^{-1} s^{n/2} \leq |S_u(z, s)| \leq C s^{n/2} \quad \text{if } S_u(z, s) \Subset \Omega,$$

where $C = C(n, \lambda, \Lambda)$, and consequences of [26, Lemma 5.6(ii) and Corollary 5.22]:

$$(2.5) \quad [Du]_{C^\alpha(\bar{S}_{2h_0})} \leq M, \quad \text{dist}(\bar{S}_{3h_0/2}, \partial S_{2h_0}) \geq d_0 > 0 \quad \text{if } S_{4h_0} = S_u(x_0, 4h_0) \Subset \Omega.$$

Here M and d_0 depend only on n, λ, Λ, h_0 and $\text{diam}(S_{4h_0})$. Specifically, in [26], Corollary 5.22 can be applied at height $3h_0$ with $\tau = 4/3$ and $\delta = 2/3$; Lemma 5.6(ii), at height $2h_0$ and ratio $3/4$, gives $d_0 \geq C^{-1} h_0^{n/2} [\text{diam}(S_{4h_0})]^{1-n}$.

We recall a localization property of superlevel sets of the Green's function of the linearized Monge-Ampère operator that will be needed in the proof of Proposition 1.2.

Lemma 2.6. *Let $u \in C^3(\Omega)$ be a convex function satisfying (1.1), where $\Omega \subset \mathbb{R}^n$ ($n \geq 3$). Assume $S_u(z, 2h) \Subset \Omega$ where $z \in \Omega$ and $h > 0$. Let $g_S(\cdot, y)$ be the Green's function of the linearized Monge-Ampère operator $D_i(U^{ij} D_j)$ in $S = S_u(z, h)$ with pole $y \in S$.*

(i) *(Localization of superlevel sets) There are constants $c_l = c_l(n, \lambda, \Lambda) > 0$ and $\tau = \tau(n, \lambda, \Lambda) > 1$ such that, for every $y \in S$ and $k \geq \tau h^{-a}$,*

$$(2.6) \quad \{x \in S : g_S(x, y) > k\} \subset S_u(y, c_l k^{-1/a}) \subset S_u(y, 2c_l k^{-1/a}) \Subset S_u(z, 3h/2).$$

(ii) *(Truncation identity) Let $y \in S$. Then, for any $k > 0$, we have the truncation identity (see [26, (14.23), equivalently (14.71)])*

$$(2.7) \quad \int_{\{g_S(\cdot, y) \leq k\}} U^{ij} D_i g_S(\cdot, y) D_j g_S(\cdot, y) dx = k.$$

Proof. We indicate the derivation of (2.6) for the reader's convenience. Indeed, [26, Theorem 5.13(iii)], applied in $S_u(z, 2h)$, gives a constant $\eta = \eta(n, \lambda, \Lambda) > 0$ such that $S_u(y, \eta h) \Subset S_u(z, 3h/2)$ for all $y \in S$. If $k \geq \tau h^{-a}$ where $\tau = \tau(n, \lambda, \Lambda) > 1$ is large, then [16, Lemma 4.1(i)] gives the first inclusion in (2.6) with a constant $c_l = c_l(n, \lambda, \Lambda) > 0$. Increasing τ ensures that $2c_l k^{-1/a} < \eta h/2$, and this gives the last inclusion, as desired. $\square$

Remark 2.7 (Euclidean gradient versus Monge–Ampère gradient). Let $u \in C^3(\Omega)$ be a convex function satisfying $\det D^2 u \geq \lambda > 0$. By [26, Lemma 2.58(iii)], we have for all function v ,

$$(2.8) \quad |Dv|^2 \leq \lambda^{-1}(\Delta u) U^{ij} D_i v D_j v \equiv \lambda^{-1}(\Delta u)(\det D^2 u)^{-1/2} |Dv|^2.$$

3. THE GREEN'S FUNCTION GRADIENT ESTIMATE

In this section, we prove Proposition 1.2 and its extension to higher integrability. Unless otherwise indicated, we assume $n \geq 3$, (1.1) and (1.3), so

$$0 < \lambda \leq \det D^2 u \leq \Lambda \quad \text{in } \Omega \subset \mathbb{R}^n, \quad S_{4h_0} = S_u(x_0, 4h_0) \Subset \Omega.$$

3.1. Integrability of the Green gradient. The following elementary consequence of (2.5) provides the bound of the mass of the Hessian used in the Green's function estimate.

Lemma 3.1 (Hessian mass on a section). *Assume $n \geq 3$, (1.1) and (1.3). For every $S_u(z, s) \Subset S_{3h_0/2}$, we have*

$$(3.1) \quad \int_{S_u(z, s)} \Delta u \, dx \leq C s^{a+2\beta},$$

where a and β are defined as in (2.3), and $C = C(n, \lambda, \Lambda, h_0, \text{diam}(S_{4h_0}))$.

Proof. Let $S := S_u(z, s)$. By subtracting $u(z) + Du(z) \cdot (x - z)$ from u , the claimed estimate remains the same, but we can now assume $u \geq 0$, $u(z) = 0$, and $u = s$ on ∂S .

For $x \in \partial S$, set $e = Du(x)/|Du(x)|$ and $r = cs^{1/(1+\alpha)}$. The gradient does not vanish on this positive level set. From $S_u(z, s) \Subset S_{3h_0/2}$ and (2.4), we have $s \leq C_1 h_0$ where $C_1 = C_1(n, \lambda, \Lambda) > 0$. Thus, there exists $c = c(n, \lambda, \Lambda, h_0) > 0$ such that $r \leq h_0/2$, so the segment $\{x - te : 0 \leq t \leq r\} \subset S_{2h_0}$. By (2.5),

$$0 \leq u(x - re) \leq s - r|Du(x)| + \frac{M}{1+\alpha} r^{1+\alpha}.$$

Consequently, we can find $C_2 = C_2(n, \lambda, \Lambda, h_0, \text{diam}(S_{4h_0}))$ such that

$$(3.2) \quad m_s := \sup_{\partial S} |Du| \leq C_2 s^\beta.$$

On ∂S , the outward normal is $\nu = Du/|Du|$, and convexity gives $Du(x) \cdot (x - z) \geq s$. The divergence theorem therefore yields

$$s \int_S \Delta u \, dx = s \int_{\partial S} |Du| \, d\mathcal{H}^{n-1} \leq \int_{\partial S} |Du|^2 (x - z) \cdot \nu \, d\mathcal{H}^{n-1} \leq n|S|m_s^2.$$

Combining this with (2.4) and (3.2) proves (3.1), since $n/2 - 1 + 2\beta = a + 2\beta$. $\square$

We now establish the L^1 estimate for the gradient of the Green's function.

Proof of Proposition 1.2. In this proof, the constants C depend only on n, λ, Λ, h_0 , and $\text{diam}(S_{4h_0})$. Fix $y \in S$. Then, from (2.8), for every measurable $E \subset S$,

$$(3.3) \quad \int_E |Dg| \, dx \leq \lambda^{-1/2} \left(\int_E U^{ij} D_i g D_j g \, dx \right)^{1/2} \left(\int_E \Delta u \, dx \right)^{1/2}.$$

Let τ be as in (2.6) and $T = \tau h^{-a}$. Equations (2.7) and (3.1) imply

$$(3.4) \quad \int_{\{g \leq T\}} |Dg| \, dx \leq CT^{1/2} h^{(a+2\beta)/2} \leq Ch^\beta.$$

For $j \geq 0$, let $E_j = \{2^j T < g \leq 2^{j+1} T\}$. Then by (3.1), (2.6) and (2.7), we have

$$\int_{E_j} U^{ik} D_i g D_k g \, dx \leq 2^{j+1} T, \quad \int_{E_j} \Delta u \, dx \leq \int_{S_u(y, c_l(2^j T)^{-1/a})} \Delta u \, dx \leq C(2^j T)^{-(a+2\beta)/a}.$$

The last two inequalities also use $\Delta u \geq 0$. By (3.3),

$$\int_{E_j} |Dg| \, dx \leq C(2^j T)^{-\beta/a} \leq Ch^\beta 2^{-j\beta/a}.$$

Since $a, \beta > 0$, summing over j and adding (3.4) proves (1.5), with a constant independent of y . The proposition is proved. $\square$

We show that solutions to the linearized Monge–Ampère equation with right-hand side being the divergence of a bounded vector field and vanish on the boundary of a section decay to zero when the section shrinks to a point.

Corollary 3.2. *Assume $n \geq 3$, (1.1) and (1.3). Let $S = S_u(z, h)$ with $S_u(z, 2h) \Subset S_{3h_0/2}$, and let $\mathbf{F} \in L^\infty(S; \mathbb{R}^n)$. The weak solution $w \in W_0^{1,2}(S)$ of*

$$L_u w = \text{div } \mathbf{F} \quad \text{in } S, \quad w = 0 \quad \text{on } \partial S$$

satisfies

$$(3.5) \quad \|w\|_{L^\infty(S)} \leq Ch^\beta \|\mathbf{F}\|_{L^\infty(S)}, \quad C = C(n, \lambda, \Lambda, h_0, \text{diam}(S_{4h_0})).$$

Proof. First, consider the case where $\mathbf{F}$ is a smooth vector field. Then, the representation formula (2.2) can be applied to w , and integration by parts gives

$$w(y) = - \int_S g_S(x, y) \text{div } \mathbf{F}(x) \, dx = \int_S D_x g_S(x, y) \cdot \mathbf{F}(x) \, dx.$$

Thus, (3.5) follows from Proposition 1.2.

For a general bounded vector field $\mathbf{F}$, we extend it by zero and mollify to obtain smooth vector fields $\mathbf{F}_k \rightarrow \mathbf{F}$ in $L^2(S)$ with $\|\mathbf{F}_k\|_{L^\infty(S)} \leq \|\mathbf{F}\|_{L^\infty(S)}$. The corresponding weak solutions $w_k \in W_0^{1,2}(S)$ converge to w in $W_0^{1,2}(S)$ by the energy estimate for their difference. Here, the existence and convergence use the ellipticity of the fixed smooth matrix U on $\bar{S}$. Moreover, there is a subsequence $\{w_{k_l}\}$ that converges almost everywhere to w . Thus, letting $l \rightarrow \infty$ in $\|w_{k_l}\|_{L^\infty(S)} \leq Ch^\beta \|\mathbf{F}_{k_l}\|_{L^\infty(S)}$ proves (3.5). $\square$

3.2. Higher integrability of the gradient of the Green's function. The proof of Proposition 1.2 deliberately avoids any higher integrability input for D^2u . If one now invokes the classical interior $W^{2,1+\varepsilon}$ estimate for the Monge–Ampère equation of De Philippis–Figalli–Savin [12] and, independently, Schmidt [37], the same level-band localization yields a genuine improvement of the integrability of the gradient of the Green's function in every dimension $n \geq 3$. The result of this section is not used in the proof of Theorem 1.1.

We first record a consequence of the interior $W^{2,1+\varepsilon}$ estimate that will be used below.

Lemma 3.3. *Assume $n \geq 3$, (1.1) and (1.3). There exists $\varepsilon_0 = \varepsilon_0(n, \lambda, \Lambda) > 0$ such that, for every $\delta \in (0, \varepsilon_0]$, the following holds. If $S_u(\xi, 2s) \in S_{3h_0/2}$, then*

$$(3.6) \quad \int_{S_u(\xi, s)} (\Delta u)^{1+\delta} dx \leq C_\delta s^{a+2\beta-\delta(1-2\beta)}, \quad C_\delta = C_\delta(n, \lambda, \Lambda, h_0, \text{diam}(S_{4h_0})).$$

Proof. From the interior $W^{2,1+\varepsilon_0}$ estimate for the Monge–Ampère equation [12, 37], we have (see also [34, Equation (1.2) and Lemma 3.1]) for some $C_0 = C_0(n, \lambda, \Lambda)$,

$$\begin{aligned} \int_{S_u(\xi, s)} (\Delta u)^{1+\varepsilon_0} dx &\leq C_0 |S_u(\xi, s)|^{-\varepsilon_0} \left(\int_{S_u(\xi, s)} \Delta u dx \right)^{1+\varepsilon_0} \\ &\leq C s^{(1+\varepsilon_0)(a+2\beta)-\frac{n}{2}\varepsilon_0} \\ &= C s^{a+2\beta-\varepsilon_0(1-2\beta)}, \quad C = C(n, \lambda, \Lambda, h_0, \text{diam}(S_{4h_0})), \end{aligned}$$

where we used Lemma 3.1 and the volume estimates (2.4) for sections. Note that

$$a + 2\beta - \frac{n}{2} = 2\beta - 1.$$

The claimed estimate (3.6) holds for $\delta = \varepsilon_0$. For $0 < \delta < \varepsilon_0$, an application of the Hölder inequality and the volume estimates for sections give the desired result. $\square$

Proposition 3.4 (Higher integrability of the gradient of the Green's function). *Let $n \geq 3$, and assume (1.1) and (1.3). Let $S = S_u(z, h)$ satisfy $S_u(z, 2h) \in S_{3h_0/2}$. Then there exists $\varepsilon = \varepsilon(n, \lambda, \Lambda) > 0$ such that, with*

$$(3.7) \quad \vartheta = (1 + \varepsilon)\beta - \frac{n}{2}\varepsilon > 0,$$

one has

$$(3.8) \quad \sup_{y \in S} \int_S |D_x g_S(x, y)|^{1+\varepsilon} dx \leq C h^\vartheta, \quad C = C(n, \lambda, \Lambda, h_0, \text{diam}(S_{4h_0})).$$

In particular, $g_S(\cdot, y) \in W_0^{1,1+\varepsilon}(S)$, uniformly in the pole.

Proof. In this proof, the constants C depend only on n, λ, Λ, h_0 , and $\text{diam}(S_{4h_0})$. Let ε_0 be given by Lemma 3.3. Choose

$$(3.9) \quad 0 < \varepsilon = \varepsilon(n, \lambda, \Lambda) < \min \left\{ \frac{\varepsilon_0}{2 + \varepsilon_0}, \frac{2\beta}{n - 2\beta} \right\},$$

and set

$$(3.10) \quad \delta = \frac{2\varepsilon}{1 - \varepsilon}.$$

Then $0 < \delta < \varepsilon_0$,

$$1 + \delta = \frac{1 + \varepsilon}{1 - \varepsilon},$$

and (3.7) follows from the second inequality in (3.9).

Fix $y \in S$ and write $g := g_S(\cdot, y)$. Then, Hölder's inequality with exponents $2/(1 + \varepsilon)$ and $2/(1 - \varepsilon)$ applied to (2.8) gives, for every measurable $E \subset S$,

$$(3.11) \quad \int_E |Dg|^{1+\varepsilon} dx \leq C_1(n, \lambda, \Lambda) \left(\int_E U^{ij} D_i g D_j g dx \right)^{\frac{1+\varepsilon}{2}} \left(\int_E (\Delta u)^{1+\delta} dx \right)^{\frac{1-\varepsilon}{2}}.$$

Let $T = \tau h^{-a}$. We recall (2.6), in particular,

$$(3.12) \quad S_u(y, 2c_l k^{-1/a}) \Subset S_u(z, 3h/2) \quad \text{for all } k \geq T.$$

For the sublevel set $\{g \leq T\}$, (2.7), Lemma 3.3, and (3.11) yield

$$\int_{\{g \leq T\}} |Dg|^{1+\varepsilon} dx \leq C_1 T^{\frac{1+\varepsilon}{2}} \left(\int_S (\Delta u)^{1+\delta} dx \right)^{\frac{1-\varepsilon}{2}} \leq Ch^\vartheta.$$

Here and below we use $a + 1 = n/2$ and (3.10); explicitly,

$$(3.13) \quad -\frac{a(1 + \varepsilon)}{2} + \frac{1 - \varepsilon}{2} [a + 2\beta - \delta(1 - 2\beta)] = \vartheta.$$

For $j \geq 0$, set

$$K_j = 2^j T, \quad E_j = \{K_j < g \leq 2K_j\}, \quad s_j = c_l K_j^{-1/a}.$$

By (2.7),

$$\int_{E_j} U^{ik} D_i g D_k g dx \leq 2K_j.$$

Moreover, (2.6) and (3.12) give

$$E_j \subset S_u(y, s_j), \quad S_u(y, 2s_j) \Subset S_{3h_0/2}.$$

Applying Lemma 3.3 and (3.11), and then using $K_j \approx s_j^{-a}$ (within a positive factor depending only on n, λ , and Λ) together with (3.13), we obtain

$$\int_{E_j} |Dg|^{1+\varepsilon} dx \leq C s_j^\vartheta = Ch^\vartheta 2^{-j\vartheta/a}.$$

Recalling $\vartheta > 0$, we find

$$\begin{aligned} \int_S |Dg|^{1+\varepsilon} dx &= \int_{\{g \leq T\}} |Dg|^{1+\varepsilon} dx + \sum_{j=0}^{\infty} \int_{E_j} |Dg|^{1+\varepsilon} dx \\ &\leq Ch^\vartheta (1 + \sum_{j=0}^{\infty} 2^{-j\vartheta/a}) \leq Ch^\vartheta. \end{aligned}$$

Thus (3.8) is established, with a constant independent of y .

Finally, combining (3.8) with the L^q estimate for the Green's function in [26, Theorem 14.11] yields $g \in W_0^{1,1+\varepsilon}(S)$. $\square$

4. INTERIOR HÖLDER ESTIMATE

We prove interior Hölder estimates for the linearized Monge–Ampère equation in divergence form in this section. Unless otherwise indicated, we assume $n \geq 3$, (1.1) and (1.3).

4.1. Bounded vector fields. For the fixed smooth strictly convex potential function u , it is known (see the argument in [26, Remark 14.3]) that $W^{1,2}$ weak solutions to the homogeneous linearized Monge–Ampère equation

$$U^{ij} D_{ij} w = 0 \quad \text{in} \quad S_u(z, r) \Subset \Omega$$

are locally in $W^{2,n}$. We use two standard estimates.

From the Caffarelli–Gutiérrez Harnack inequality [26, Theorem 12.4], we have the following oscillation estimate (see [26, (12.26)])

$$(4.1) \quad \text{osc}_{S_u(z, r/2)} w \leq \theta \text{osc}_{S_u(z, r)} w, \quad 0 < \theta = \theta(n, \lambda, \Lambda) < 1.$$

Moreover, the nonnegative subsolution estimate in [6, Lemma 2.4], applied to $|w|$, gives

$$(4.2) \quad \sup_{S_u(z, r/2)} |w| \leq C_p \left(\int_{S_u(z, r)} |w|^p dx \right)^{1/p}, \quad C_p = C_p(n, \lambda, \Lambda, p).$$

From the inclusion property of interior sections [26, Theorem 5.30(i)], we can find a constant $c_* = c_*(n, \lambda, \Lambda) > 0$ such that as long as $S_{4h_0} = S_u(x_0, 4h_0) \Subset \Omega$, $r_* = c_* h_0$ satisfies

$$(4.3) \quad S_u(z, 4r_*) \Subset S_{3h_0/2} \quad \text{and} \quad S_u(x, 4r_*) \Subset S_{5h_0/4}$$

for all $z \in S_{5h_0/4}$ and $x \in S_{h_0}$.

Proof of Theorem 1.1. In this proof, the constants C depend only on n, λ, Λ, h_0 , and $\text{diam}(S_{4h_0})$. Write

$$F_0 = \|\mathbf{F}\|_{L^\infty(S_{2h_0})} \quad \text{and} \quad V_p = \|v\|_{L^p(S_{2h_0})}.$$

For a fixed, strictly convex function $u \in C^3(\Omega)$, the classical local elliptic regularity theory gives continuous representatives of the solutions used below; see [15, Chapter 8]. This observation is qualitative only. The constants in the following estimates are structural.

Step 1: Local boundedness. For $z \in S_{5h_0/4}$, let $w_z \in W_0^{1,2}(S_u(z, 2r_*))$ be the solution to

$$L_u w_z = \text{div } \mathbf{F} \quad \text{in } S_u(z, 2r_*), \quad w_z = 0 \quad \text{on } \partial S_u(z, 2r_*).$$

By (4.3) and Corollary 3.2,

$$\|w_z\|_{L^\infty(S_u(z, 2r_*))} \leq C h_0^\beta F_0.$$

The function $H_z := v - w_z$ satisfies the homogeneous equation

$$L_u H_z = 0 \quad \text{in } S_u(z, 2r_*).$$

Applying (4.2) on $S_u(z, r_*)$ and using (2.4), we obtain

$$|v(z)| \leq C_p |S_u(z, r_*)|^{-1/p} \|H_z\|_{L^p(S_u(z, r_*))} + \|w_z\|_{L^\infty(S_u(z, r_*))} \leq C_p (h_0^{-n/(2p)} V_p + h_0^\beta F_0).$$

Consequently,

$$(4.4) \quad \|v\|_{L^\infty(S_{5h_0/4})} \leq C_p (h_0^{-n/(2p)} V_p + h_0^\beta F_0).$$

Step 2: Oscillation decay. Fix $x \in S_{h_0}$ and $0 < h \leq r_*$. On $S = S_u(x, h)$, split

$$v = H + w,$$

where

$$L_u w = \operatorname{div} \mathbf{F} \quad \text{in } S, \quad w = 0 \quad \text{on } \partial S,$$

and

$$L_u H = 0 \quad \text{in } S, \quad H = v \quad \text{on } \partial S.$$

By the weak maximum principle,

$$\operatorname{osc}_S H \leq \operatorname{osc}_S v.$$

From (4.1) (applied to H) and (3.5) in Corollary 3.2, we have

$$(4.5) \quad \operatorname{osc}_{S_u(x, h/2)} v \leq \operatorname{osc}_{S_u(x, h/2)} H + \operatorname{osc}_{S_u(x, h/2)} w \leq \theta \operatorname{osc}_{S_u(x, h)} v + Ch^\beta F_0.$$

This is the splitting argument in the proof of [26, Theorem 15.6], with Corollary 3.2 supplying the uniform estimate for zero-boundary solutions. Choose

$$\kappa = \frac{1}{2} \min\{\beta, -\log_2 \theta\} > 0.$$

The geometric iteration used in [26, proof of Theorem 12.13] gives

$$(4.6) \quad \operatorname{osc}_{S_u(x, h)} v \leq C \left(\frac{h}{r_*} \right)^\kappa (\operatorname{osc}_{S_u(x, r_*)} v + r_*^\beta F_0), \quad 0 < h \leq r_*.$$

Note that, equations (4.3) and (4.4) bound the initial oscillation in (4.6)

$$\operatorname{osc}_{S_u(x, r_*)} v \leq C_p(V_p + F_0).$$

Step 3: Hölder continuity. By (2.5), for $x, y \in S_{h_0}$,

$$0 \leq u(y) - u(x) - Du(x) \cdot (y - x) \leq B|x - y|^{1+\alpha}, \quad B = B(n, \lambda, \Lambda, h_0, \operatorname{diam}(S_{4h_0})).$$

If $x \neq y$ and $h = 2B|x - y|^{1+\alpha} \leq r_*$, then $y \in S_u(x, h)$, and (4.6) yields

$$|v(x) - v(y)| \leq C_p(V_p + F_0)|x - y|^\gamma, \quad \gamma = (1 + \alpha)\kappa.$$

If $h = 2B|x - y|^{1+\alpha} \geq r_*$, then $|x - y| \geq (r_0/B)^{\frac{1}{1+\alpha}}$, so the same estimate follows from (4.4). Since $0 < \gamma \leq \alpha/2 < 1$ and $\gamma = \gamma(n, \lambda, \Lambda)$, this proves (1.4). $\square$

Remark 4.1. The proof does not use a higher Lebesgue norm of D^2u . Its new input is Proposition 1.2.

4.2. L^q vector fields. This section records a possible strengthening of Theorem 1.1 that follows from Proposition 3.4. The point is that the gain from $D_x g_S \in L^{1+\varepsilon}$ allows the boundedness assumption on the divergence datum to be replaced by sufficiently high, but finite, Lebesgue integrability.

Fix once and for all an exponent $\varepsilon_G > 0$ for which Proposition 3.4 holds, and set

$$q_* := \frac{1 + \varepsilon_G}{\varepsilon_G}, \quad \vartheta_G := (1 + \varepsilon_G)\beta - \frac{n}{2}\varepsilon_G > 0.$$

Then

$$(4.7) \quad \frac{\vartheta_G}{1 + \varepsilon_G} = \beta - \frac{n}{2q_*} > 0.$$

In particular, $q_* > n/(2\beta) > n$ because $\beta < 1/2$.

Proposition 4.2 (Uniform estimate for zero-boundary solutions with L^q vector fields). *Assume $n \geq 3$, (1.1) and (1.3). Let $S = S_u(z, h)$ satisfy $S_u(z, 2h) \Subset S_{3h_0/2}$, and let $q \in [q_*, \infty)$. If $\mathbf{F} \in L^q(S; \mathbb{R}^n)$ and $w \in W_0^{1,2}(S)$ solves*

$$L_u w = \operatorname{div} \mathbf{F} \quad \text{in } S, \quad w = 0 \quad \text{on } \partial S,$$

then

$$(4.8) \quad \|w\|_{L^\infty(S)} \leq C_q h^{\eta_q} \|\mathbf{F}\|_{L^q(S)}, \quad \eta_q := \beta - \frac{n}{2q} > 0,$$

where $C_q = C(n, q, \lambda, \Lambda, h_0, \operatorname{diam}(S_{4h_0}))$.

Proof. Assume first that the vector field $\mathbf{F}$ is smooth. For $y \in S$, the representation formula (2.2) can be applied to w , and integration by parts gives

$$w(y) = - \int_S g_S(x, y) \operatorname{div} \mathbf{F}(x) \, dx = \int_S D_x g_S(x, y) \cdot \mathbf{F}(x) \, dx.$$

The Hölder inequality with exponents $1 + \varepsilon_G$ and q_* and Proposition 3.4 give

$$|w(y)| \leq \|D_x g_S(\cdot, y)\|_{L^{1+\varepsilon_G}(S)} \|\mathbf{F}\|_{L^{q_*}(S)} \leq C \|\mathbf{F}\|_{L^{q_*}(S)} h^{\vartheta_G/(1+\varepsilon_G)}.$$

If $q \geq q_*$, the volume estimate for sections in (2.4) yields

$$\|\mathbf{F}\|_{L^{q_*}(S)} \leq |S|^{\frac{1}{q_*} - \frac{1}{q}} \|\mathbf{F}\|_{L^q(S)} \leq C h^{\frac{n}{2}(\frac{1}{q_*} - \frac{1}{q})} \|\mathbf{F}\|_{L^q(S)}.$$

Combining these estimates and recalling (4.7), we obtain (4.8) for smooth data.

For general vector field $\mathbf{F} \in L^q(S)$, one may extend $\mathbf{F}$ by zero and mollify to obtain smooth approximants $\mathbf{F}_k$. Since $q \geq q_* > n > 2$, $\mathbf{F}_k$ converges to $\mathbf{F}$ in $L^2(S)$, so the corresponding solution to

$$L_u w_k = \operatorname{div} \mathbf{F}_k \quad \text{in } S, \quad w_k = 0 \quad \text{on } \partial S,$$

converges to w in $W_0^{1,2}(S)$ by the usual energy estimate. The uniform $L^\infty(S)$ bound on w_k passes to an almost-everywhere convergent subsequence, which proves (4.8) for w . $\square$

Theorem 4.3 (Interior Hölder estimate with L^q vector fields). *Let $n \geq 3$, and assume (1.1) and (1.3). Let $p \in (1, \infty)$ and $q \in [q_*, \infty)$. Suppose that $\mathbf{F} \in L^q(S_{2h_0}; \mathbb{R}^n)$ and $v \in W_{\operatorname{loc}}^{1,2}(S_{2h_0}) \cap L^p(S_{2h_0})$ satisfies*

$$L_u v = \operatorname{div} \mathbf{F} \quad \text{weakly in } S_{2h_0}.$$

Then there exist $\gamma_q = \gamma_q(n, q, \lambda, \Lambda) \in (0, 1)$ and $C = C(n, p, q, \lambda, \Lambda, h_0, \operatorname{diam}(S_{4h_0})) > 0$ such that

$$(4.9) \quad \|v\|_{L^\infty(S_{h_0})} + [v]_{C^{\gamma_q}(S_{h_0})} \leq C \left(\|\mathbf{F}\|_{L^q(S_{2h_0})} + \|v\|_{L^p(S_{2h_0})} \right).$$

Proof. The proof is similar to the proof of Theorem 1.1 where Corollary 3.2 is now replaced by Proposition 4.2. We include the details to keep the dependence on q and the section scale explicit. We use the estimates (4.1)-(4.2) and inclusions (4.3) as in the last subsection.

In this proof, the constants C, C_p depend only on $n, \lambda, \Lambda, p, q, h_0$, and $\operatorname{diam}(S_{4h_0})$. Write

$$F_q := \|\mathbf{F}\|_{L^q(S_{2h_0})}, \quad V_p := \|v\|_{L^p(S_{2h_0})}.$$

For $z \in S_{5h_0/4}$, let $w_z \in W_0^{1,2}(S_u(z, 2r_*))$ be the solution to

$$L_u w_z = \operatorname{div} \mathbf{F} \quad \text{in } S_u(z, 2r_*), \quad w_z = 0 \quad \text{on } \partial S_u(z, 2r_*).$$

By Proposition 4.2,

$$\|w_z\|_{L^\infty(S_u(z, 2r_*))} \leq Cr_*^{\eta_q} F_q.$$

The function $H_z := v - w_z$ satisfies the homogeneous equation

$$L_u H_z = 0 \quad \text{in } S_u(z, 2r_*).$$

Applying (4.2) on $S_u(z, r_*)$ and using (2.4),

$$|v(z)| \leq C_p |S_u(z, r_*)|^{-1/p} \|H_z\|_{L^p(S_u(z, r_*))} + \|w_z\|_{L^\infty(S_u(z, r_*))} \leq C(V_p + F_q),$$

where $r_*^{-n/(2p)}$ is absorbed into the constant C . Consequently,

$$(4.10) \quad \|v\|_{L^\infty(S_{5h_0/4})} \leq C(V_p + F_q).$$

Next fix $x \in S_{h_0}$ and $0 < h \leq r_*$. Then, as in the derivation of (4.5) where Corollary 3.2 is now replaced by Proposition 4.2, we have

$$(4.11) \quad \text{osc}_{S_u(x, h/2)} v \leq \theta \text{osc}_{S_u(x, h)} v + Ch^{\eta_q} F_q.$$

Since $\eta_q > 0$, choose

$$\kappa_q := \frac{1}{2} \min\{\eta_q, -\log_2 \theta\} > 0.$$

The standard geometric iteration then yields

$$(4.12) \quad \text{osc}_{S_u(x, h)} v \leq C \left(\frac{h}{r_*} \right)^{\kappa_q} \left(\text{osc}_{S_u(x, r_*)} v + r_*^{\eta_q} F_q \right), \quad 0 < h \leq r_*.$$

By (4.10), the quantity in parentheses is bounded by $C(V_p + F_q)$.

As in Step 3 in the proof of Theorem 1.1, we have (4.9) with $\gamma_q = (1 + \alpha)\kappa_q \in (0, 1)$. Indeed, $\eta_q \leq \beta$, hence $\gamma_q \leq (1 + \alpha)\beta/2 = \alpha/2 < 1$. $\square$

Remark 4.4. The above extension of the interior Hölder estimate in Theorem 1.1 to L^q vector fields uses the higher-integrability theory for the Hessian D^2u through Proposition 3.4. Thus, it is logically separate from Theorem 1.1, whose proof for bounded vector fields does not use interior $W^{2,1+\varepsilon}$ estimate of the Monge–Ampère equation.

5. GLOBAL HÖLDER ESTIMATES

We study global Hölder estimates for the linearized Monge–Ampère equation in divergence form in this section under the following structural conditions.

Global structural assumptions. Let $\Omega \subset \mathbb{R}^n$ be a convex domain and assume that there exists a constant $\rho > 0$ such that

$$(5.1) \quad \Omega \subset B_{1/\rho}(0) \subset \mathbb{R}^n$$

and for each $y \in \partial\Omega$,

$$(5.2) \quad \text{there is an interior ball } B_\rho(z) \subset \Omega \text{ such that } y \in \partial B_\rho(z).$$

Let $u \in C^{1,1}(\overline{\Omega}) \cap C^3(\Omega)$ be a convex function satisfying

$$(5.3) \quad 0 < \lambda \leq \det D^2u \leq \Lambda \quad \text{in } \Omega.$$

Assume further that on $\partial\Omega$, u separates quadratically from its tangent hyperplanes; namely, for all $x_0, x \in \partial\Omega$, we have

$$(5.4) \quad \rho |x - x_0|^2 \leq u(x) - u(x_0) - Du(x_0) \cdot (x - x_0) \leq \rho^{-1} |x - x_0|^2.$$

For the boundary localization framework associated with this hypothesis, see Savin [36].

5.1. Global integrability of the gradient of the Green's function. We first recall [16, Lemma 3.5].

Lemma 5.1. *Assume that u and Ω satisfy (5.1)–(5.4). There exists $\alpha_b = \alpha_b(n, \lambda, \Lambda) \in (0, 1)$ such that the following statements hold.*

(i) *There exists $C_* = C_*(n, \lambda, \Lambda, \rho) > 0$ such that*

$$[Du]_{C^{\alpha_b}(\bar{\Omega})} := \sup_{x \neq y \in \bar{\Omega}} |Du(x) - Du(y)|/|x - y|^{\alpha_b} \leq C_*.$$

(ii) *Let $S_u(x_0, t_0)$ be a section of u with $x_0 \in \bar{\Omega}$ and $t_0 > 0$. Then*

$$\mathcal{H}^{n-1}(\partial S_u(x_0, t_0)) \leq C(n, \lambda, \Lambda, \rho) t_0^{\frac{n-2}{2} + \frac{\alpha_b}{\alpha_b+1}}.$$

The following lemma is a global version of Proposition 1.2.

Lemma 5.2 (Hessian mass). *Assume that u and Ω satisfy (5.1)–(5.4). Let $S_u(x_0, t_0)$ be a section of u with $x_0 \in \bar{\Omega}$ and $t_0 > 0$. Then*

$$\int_{S_u(x_0, t_0)} \Delta u \, dx \leq C t_0^{\frac{n-2}{2} + \frac{\alpha_b}{\alpha_b+1}}, \quad C = C(n, \lambda, \Lambda, \rho).$$

Proof. By the divergence theorem, we have

$$\begin{aligned} \int_{S_u(x_0, t_0)} \Delta u \, dx &= \int_{S_u(x_0, t_0)} \Delta [u(x) - u(x_0) - Du(x_0) \cdot (x - x_0)] \, dx \\ &= \int_{\partial S_u(x_0, t_0)} [Du(x) - Du(x_0)] \cdot \nu \, dS. \end{aligned}$$

The claimed estimate now follows from Lemma 5.1, noting that

$$|Du(x) - Du(x_0)| \leq C_* |x - x_0|^{\alpha_b} \leq C_*(2/\rho)^{\alpha_b}, \quad \text{for all } x \in \partial S_u(x_0, t_0).$$

□

Remark 5.3. Observe that, under the assumptions (5.1)–(5.4), by [26, Lemma 9.7], there exists $M = M(n, \lambda, \Lambda, \rho) > 0$ such that

$$\bar{\Omega} \subset S_u(x_0, M) \quad \text{for all } x_0 \in \bar{\Omega}.$$

Therefore, Ω is a section of u with height comparable to 1. Consider an open, proper subset $V \subset \Omega$. By the global localization of the superlevel sets of the Green's function of the Monge–Ampère operator in [16, Lemma 4.2(i)], there exist constants $C_g, T_g > 0$ depending only on $n, \lambda, \Lambda, \rho$ such that, for every $y \in V$, and every $k \geq T_g$,

$$(5.5) \quad \{x \in V : g_V(x, y) > k\} \subset S_u(y, C_g k^{-\frac{2}{n-2}}).$$

Moreover, if $g = g_V(\cdot, y)$, then

$$(5.6) \quad |Dg|^2 \leq \lambda^{-1} \Delta u U^{ij} D_i g D_j g, \quad \int_{\{g \leq k\}} U^{ij} D_i g D_j g \, dx = k.$$

As in the proof of Proposition 1.2 (without using the global $W^{2,1+\varepsilon}$ -estimate for the Monge–Ampère equation), we have

Proposition 5.4. *Let $n \geq 3$, and assume that u and Ω satisfy (5.1)–(5.4). Let $S := S_u(x_0, t_0)$ be a section of u with $x_0 \in \bar{\Omega}$ and $t_0 > 0$. Then*

$$\sup_{y \in S} \int_S |D_x g_S(x, y)| dx \leq C t_0^{\frac{\alpha_b}{2(\alpha_b+1)}}, \quad C = C(n, \lambda, \Lambda, \rho).$$

In particular,

$$\sup_{y \in \Omega} \int_{\Omega} |D_x g_{\Omega}(x, y)| dx \leq C(n, \lambda, \Lambda, \rho).$$

Throughout Section 5, we assume the global structural hypotheses (5.1)–(5.4), and write

$$a = \frac{n-2}{2}, \quad \beta_{\partial} := \frac{\alpha_b}{1+\alpha_b},$$

where α_b is the exponent in Lemma 5.1.

We first upgrade the global L^1 estimate for the gradient of the Green's function. The point is that the Hessian mass decay with respect to the section height in Lemma 5.2, together with the known global higher integrability of D^2u , yields a positive gain over L^1 on every level band of the Green's function. We use the global higher-integrability estimate in [26, Theorem 10.1]; the same estimate is also recorded in the proof of [23, Theorem 2.1, Step 1] under the present structural assumptions. Thus, for some

$$(5.7) \quad \delta_0 = \delta_0(n, \lambda, \Lambda) > 0, \quad \int_{\Omega} (\Delta u)^{1+\delta_0} dx \leq C(n, \lambda, \Lambda, \rho).$$

Interpolating (5.7) with Lemma 5.2, we obtain, for every $0 < \delta < \delta_0$ and every section $S_u(\xi, s)$,

$$(5.8) \quad \int_{S_u(\xi, s)} (\Delta u)^{1+\delta} dx \leq C_{\delta} s^{(a+\beta_{\partial})(1-\delta/\delta_0)}.$$

Indeed, the interpolation parameter θ determined by

$$\frac{1}{1+\delta} = \theta + \frac{1-\theta}{1+\delta_0}$$

satisfies $\theta(1+\delta) = 1 - \delta/\delta_0$, which gives (5.8).

Proposition 5.5 (Global higher integrability of the gradient of the Green's function). *There exists $\varepsilon_b = \varepsilon_b(n, \lambda, \Lambda) > 0$ such that the following holds. If $V \subset \Omega$ is open and proper (including $V = \Omega$) and $g_V(\cdot, y)$ is the Dirichlet Green function of L_u in V , then*

$$(5.9) \quad \sup_{y \in V} \int_V |D_x g_V(x, y)|^{1+\varepsilon_b} dx \leq C(n, \lambda, \Lambda, \rho).$$

Proof. We will use (5.5) and (5.6). Let $0 < \varepsilon_b < 1/2$ to be determined with

$$\delta = \frac{2\varepsilon_b}{1-\varepsilon_b} < \delta_0.$$

Applying the Hölder inequality exactly as in (3.11) gives for every measurable $E \subset V$

$$(5.10) \quad \int_E |Dg|^{1+\varepsilon_b} dx \leq C \left(\int_E U^{ij} D_i g D_j g dx \right)^{\frac{1+\varepsilon_b}{2}} \left(\int_E (\Delta u)^{1+\delta} dx \right)^{\frac{1-\varepsilon_b}{2}}.$$

Let T_g be a fixed structural constant as in Remark 5.3 and set

$$K_j = 2^j T_g, \quad E_j = \{K_j < g \leq 2K_j\}, \quad s_j = C_g K_j^{-1/a}.$$

By (5.5), $E_j \subset S_u(y, s_j)$, while (5.6) gives $\int_{E_j} U^{ik} D_i g D_k g \leq 2K_j$. Hence (5.8) and (5.10) yield

$$(5.11) \quad \int_{E_j} |Dg|^{1+\varepsilon_b} dx \leq C s_j^{\Theta_b},$$

where

$$\Theta_b = -\frac{a(1+\varepsilon_b)}{2} + \frac{1-\varepsilon_b}{2}(a+\beta_\partial) \left(1 - \frac{\delta}{\delta_0}\right) = \frac{\beta_\partial}{2} - \varepsilon_b \left(a + \frac{\beta_\partial}{2} + \frac{a+\beta_\partial}{\delta_0}\right).$$

Since $\Theta_b \rightarrow \beta_\partial/2 > 0$ as $\varepsilon_b \rightarrow 0^+$, we may choose $\varepsilon_b = \varepsilon_b(n, \lambda, \Lambda) > 0$ so small that both

$$\delta < \delta_0 \quad \text{and} \quad \Theta_b > 0.$$

Then $s_j \approx 2^{-j/a}$ and the right-hand side of (5.11) is summable in j .

For the sublevel set $\{g \leq T_g\}$, we use (5.6), (5.7), and (5.10) to obtain a uniform bound

$$\int_{\{g \leq T_g\}} |Dg|^{1+\varepsilon_b} dx \leq C.$$

Summing (5.11) over j and adding the resulting estimate to the latest one, one obtains (5.9) as asserted. $\square$

5.2. A global Hölder estimate with L^q vector fields. We record here a global counterpart of the uniform estimate with L^q vector fields from Section 4.2. As in that section, the argument uses higher integrability of D^2u , and is therefore logically separate from the bounded-data proof of Theorem 1.1.

Fix an $\varepsilon_b = \varepsilon_b(n, \lambda, \Lambda) > 0$ as Proposition 5.5 and define

$$q_b := \frac{1+\varepsilon_b}{\varepsilon_b}.$$

After decreasing ε_b if necessary, we may and do assume $q_b > n$. For $q > q_b$, set

$$\kappa_q := \frac{1}{q_b} - \frac{1}{q} > 0, \quad \eta_q := \frac{n}{2}\kappa_q > 0.$$

Lemma 5.6 (Maximum principle with L^q vector fields). *Assume $n \geq 3$ and that u and Ω satisfy (5.1)–(5.4). Let $V \subset \Omega$ be open and proper. Let $q > q_b$ and $\mathbf{F} \in L^q(V; \mathbb{R}^n)$. If $w \in W_0^{1,2}(V)$ solves*

$$(5.12) \quad L_u w = \operatorname{div} \mathbf{F} \quad \text{in } V, \quad w = 0 \quad \text{on } \partial V,$$

then, there exists $C = C(n, \lambda, \Lambda, \rho, q)$ such that

$$(5.13) \quad \|w\|_{L^\infty(V)} \leq C|V|^{\kappa_q} \|\mathbf{F}\|_{L^q(V)}.$$

Consequently, if $v \in W^{1,2}(V) \cap C(\overline{V})$ solves $L_u v = \operatorname{div} \mathbf{F}$ in V , then

$$(5.14) \quad \|v\|_{L^\infty(V)} \leq \|v\|_{L^\infty(\partial V)} + C|V|^{\kappa_q} \|\mathbf{F}\|_{L^q(V)}.$$

Proof. As in the proof of Proposition 4.2, it suffices to prove (5.13) for smooth vector fields $\mathbf{F}$. For $y \in S$, the representation formula (2.2) can be applied to w , and integration by parts gives

$$w(y) = - \int_V g_V(x, y) \operatorname{div} \mathbf{F}(x) dx = \int_V D_x g_V(x, y) \cdot \mathbf{F}(x) dx.$$

Applying the Hölder inequality twice and using Proposition 5.5, we find

$$\|w\|_{L^\infty(V)} \leq \sup_{y \in V} \|D_x g_V(\cdot, y)\|_{L^{1+\varepsilon_b}(V)} \|\mathbf{F}\|_{L^{q_b}(V)} \leq C \|\mathbf{F}\|_{L^{q_b}(V)} \leq C|V|^{\frac{1}{q_b} - \frac{1}{q}} \|\mathbf{F}\|_{L^q(V)}.$$

For (5.14), let $w \in W_0^{1,2}(V)$ be the solution to (5.12). Then $L_u(v - w) = 0$ in V , so the weak maximum principle gives

$$\|v - w\|_{L^\infty(V)} \leq \|v\|_{L^\infty(\partial V)}.$$

This combined with (5.13) proves (5.14). $\square$

Lemma 5.6 can be applied to $V = \Omega \cap S_u(x, h)$, and in this case, using the upper bound $|S_u(x, h)| \leq C(n, \lambda)h^{n/2}$ for the volume of sections [26, Lemma 5.8], one has

$$(5.15) \quad \|w\|_{L^\infty(\Omega \cap S_u(x, h))} \leq Ch^{\eta_q} \|\mathbf{F}\|_{L^q(S_u(x, h))},$$

Thus, the usual splitting argument gives the following interior Hölder estimate.

Lemma 5.7 (Interior Hölder estimate with L^q vector fields). *Assume $n \geq 3$ and that u and Ω satisfy (5.1)–(5.4). There exists*

$$(5.16) \quad 0 < \gamma_i < \frac{1}{2} \min\{\eta_q, -\log_2 \theta\},$$

where $\theta \in (0, 1)$ is the constant in (4.1), such that if

$$L_u v = \operatorname{div} \mathbf{F} \quad \text{in} \quad S_u(y, H) \subset \Omega,$$

then for $z_1, z_2 \in S_u(y, H/8)$,

$$(5.17) \quad |v(z_1) - v(z_2)| \leq C_* \left(\|v - c\|_{L^\infty(S_u(y, H/2))} + H^{\eta_q} \|\mathbf{F}\|_{L^q(\Omega)} \right) H^{-\gamma_i} M_0^{\gamma_i} |z_1 - z_2|^{\gamma_i},$$

for every constant c , where C_* depends only on $n, \lambda, \Lambda, \rho, q$ and, in view of Lemma 5.1,

$$M_0 := \sup_{x, x' \in \bar{\Omega}} |Du(x) - Du(x')| \leq C = C(n, \lambda, \Lambda, \rho).$$

Indeed, as in the proof of [16, Theorem 6.2], estimate (5.17) follows by repeating the oscillation iteration (4.11)–(4.12) with (5.15) in place of (4.8).

We next record the boundary estimate. Its proof is the barrier argument of [16, Proposition 7.1], with Lemma 5.6 replacing the maximum-principle estimate for the inhomogeneous term.

Proposition 5.8 (Boundary Hölder estimate). *Assume $n \geq 3$ and that u and Ω satisfy (5.1)–(5.4). Let $q > q_b$ and $\sigma \in (0, 1]$. Suppose that $\mathbf{F} \in L^q(\Omega; \mathbb{R}^n)$ and $v \in W^{1,2}(\Omega) \cap C(\bar{\Omega})$ satisfies*

$$L_u v = \operatorname{div} \mathbf{F} \quad \text{in } \Omega, \quad v = \phi \quad \text{on } \partial\Omega,$$

with $\phi \in C^\sigma(\partial\Omega)$. Set

$$\sigma_0 := \min \left\{ \sigma, \frac{n\kappa_q}{2} \right\}, \quad \sigma_1 := \frac{\sigma_0}{\sigma_0 + 3n}.$$

Then there are $r_0, C > 0$, depending only on $n, \lambda, \Lambda, \rho, q$, and σ , such that

$$(5.18) \quad |v(x) - v(x_0)| \leq C(\|\phi\|_{C^\sigma(\partial\Omega)} + \|\mathbf{F}\|_{L^q(\Omega)}) |x - x_0|^{\sigma_1}$$

whenever $x_0 \in \partial\Omega$ and $x \in \Omega \cap B_{r_0}(x_0)$.

Proof. The proof follows closely those of [26, Proposition 14.32] and [16, Proposition 7.1] using maximum principles and barriers. Since our setting is different, we sketch the details for the reader's convenience.

By (5.14) with $V = \Omega$,

$$\|v\|_{L^\infty(\Omega)} \leq C(\|\phi\|_{C^\sigma(\partial\Omega)} + \|\mathbf{F}\|_{L^q(\Omega)}), \quad C = C(n, \lambda, \Lambda, \rho, q).$$

After normalization, we may assume that

$$\|v\|_{L^\infty(\Omega)} + \|\phi\|_{C^\sigma(\partial\Omega)} + \|\mathbf{F}\|_{L^q(\Omega)} = 1.$$

Fix $x_0 \in \partial\Omega$. Without loss of generality, we assume $x_0 = 0$, $u(0) = 0$, and $Du(0) = 0$.

As in the proof of [16, Proposition 7.1], for any $\varepsilon \in (0, 1)$, let

$$h_\pm := v - v(0) \pm \varepsilon \pm \frac{6}{\delta_2^3} w_{\delta_2} \quad \text{in } A := \Omega \cap B_{\delta_2}(0),$$

where $\delta_2 \in (0, 1)$ small is to be chosen later and the function w_{δ_2} is defined by

$$w_{\delta_2}(x) = w_{\delta_2}(x', x_n) := M_{\delta_2} x_n + u(x) - \tilde{\delta}_2 |x'|^2 - \Lambda^n (\lambda \tilde{\delta}_2)^{1-n} x_n^2 \quad \text{for } x = (x', x_n) \in \bar{\Omega},$$

where

$$\tilde{\delta}_2 := \delta_2^3/2 \quad \text{and} \quad M_{\delta_2} := \Lambda^n (\lambda \tilde{\delta}_2)^{1-n}.$$

If $\delta_2 \leq \varepsilon^{1/\sigma_0}$, then one can verify that

$$L_u h_+ \leq \operatorname{div} \mathbf{F} \quad \text{and} \quad L_u h_- \geq \operatorname{div} \mathbf{F} \quad \text{in } A; \quad h_- \leq 0 \quad \text{and} \quad h_+ \geq 0 \quad \text{on } \partial A.$$

Let $z_A \in W_0^{1,2}(A)$ solve

$$L_u z_A = \operatorname{div} \mathbf{F} \quad \text{in } A, \quad z_A = 0 \quad \text{on } \partial A.$$

Applying the weak maximum principle to $h_+ - z_A$ and $h_- - z_A$, and then Lemma 5.6, gives

$$(5.19) \quad \max\{-h_+, h_-\} \leq \|z_A\|_{L^\infty(A)} \leq C|A|^{\kappa_q} \|\mathbf{F}\|_{L^q(\Omega)} \leq C\delta_2^{n\kappa_q}.$$

The formula for w_{δ_2} and the boundary estimates of u give (see [16, inequality (7.3)])

$$\frac{6}{\delta_2^3} w_{\delta_2}(x) \leq C\delta_2^{-3n} |x - x_0|.$$

Choose $\delta_2 = \varepsilon^{1/\sigma_0}$. Since $\sigma_0 \leq n\kappa_q/2$, the last term in (5.19) is bounded by $C\varepsilon^2$, hence by ε after restricting to sufficiently small ε . Consequently,

$$|v(x) - v(x_0)| \leq 2\varepsilon + C\varepsilon^{-3n/\sigma_0} |x - x_0|.$$

Taking $\varepsilon = |x - x_0|^{\sigma_0/(\sigma_0+3n)}$ proves (5.18). The proof is complete. $\square$

Proof of Theorem 1.4. Note that our assumption implies (5.1)–(5.4), where ρ now depends only on $n, \lambda, \Lambda, \|u\|_{C^3(\partial\Omega)}, R$, and the C^3 regularity of $\partial\Omega$; see [26, Proposition 4.7].

As in the proof of [16, Theorem 2.4], we combine Proposition 5.8, the interior Hölder estimate (5.17) in Lemma 5.7, and the geometry of maximal interior sections. We sketch the details for the reader's convenience.

For points having at least a fixed positive distance from $\partial\Omega$, the maximal interior section has a fixed positive height, and (5.17) gives a uniform interior Hölder estimate.

Consider therefore $y \in \Omega$ with $r = \text{dist}(y, \partial\Omega)$ small. Let $S_u(y, \bar{h})$ be its maximal interior section and let $y_0 \in \partial\Omega \cap \partial S_u(y, \bar{h})$. From Savin's boundary localization theorem [36, Theorem 3.1], the maximal interior section $S_u(y, \bar{h})$ has the following properties (see [16, Proposition 7.2]),

$$(5.20) \quad cr^2 \leq \bar{h} \leq Cr^2, \quad S_u(y, \bar{h}) \subset B_{Cr|\log r|}(y), \quad B_{cr^2}(y) \subset S_u(y, \bar{h}/8).$$

For small r , Proposition 5.8 therefore gives

$$(5.21) \quad \|v - v(y_0)\|_{L^\infty(S_u(y, \bar{h}/2))} \leq C(r|\log r|)^{\sigma_1} \left(\|\phi\|_{C^\sigma(\partial\Omega)} + \|\mathbf{F}\|_{L^q(\Omega)} \right).$$

In (5.16), decrease γ_i further, if necessary, so that

$$(5.22) \quad \gamma_i < \min \left\{ \frac{\sigma_1}{4}, \frac{\eta_q}{2} \right\}.$$

Applying (5.17) on the maximal section $S_u(y, \bar{h})$ with $c = v(y_0)$ and $H = \bar{h}$, we find

$$|v(z_1) - v(z_2)| \leq C(r|\log r|)^{\sigma_1} \left(\|\phi\|_{C^\sigma(\partial\Omega)} + \|\mathbf{F}\|_{L^q(\Omega)} \right) + \bar{h}^{\eta_q} \|\mathbf{F}\|_{L^q(\Omega)} \bar{h}^{-\gamma_i} M_0^{\gamma_i} |z_1 - z_2|^{\gamma_i},$$

From (5.20)–(5.22), we have

$$(r|\log r|)^{\sigma_1} \bar{h}^{-\gamma_i} + \bar{h}^{\eta_q - \gamma_i} \leq C.$$

Thus, we can find a constant $C = C(n, \lambda, \Lambda, \rho, q, \sigma)$ such that

$$(5.23) \quad |v(z_1) - v(z_2)| \leq C|z_1 - z_2|^{\gamma_i} \quad \text{for all } z_1, z_2 \in S_u(y, \bar{h}/8).$$

Now, as in Step 3 in the proof of [16, Theorem 2.4], we obtain (1.6) with, for example,

$$\gamma = \min \left\{ \gamma_i, \frac{\sigma_1}{2} \right\} > 0.$$

Here is the basic idea. If $x, y \in \bar{\Omega}$ are sufficiently close and $|x - y| < c \text{dist}(x, \partial\Omega)^2$ (where we can assume, with out loss of generality, that $\text{dist}(x, \partial\Omega) \leq \text{dist}(y, \partial\Omega)$), then (5.23) applies. Otherwise both points lie within a distance of order $C|x - y|^{1/2}$ from the boundary, and Proposition 5.8, together with the C^σ modulus of ϕ , gives

$$|v(x) - v(y)| \leq C|x - y|^{\sigma_1/2}.$$

If $x, y \in \bar{\Omega}$ are separated by a fixed distance, the global L^∞ estimate from Lemma 5.6 suffices. The theorem is proved. $\square$

6. APPLICATIONS: SEMIGEOSTROPHIC EQUATIONS AND MONGE-AMPÈRE HOMOGENIZATION

This section presents two applications of our Hölder estimates.

6.1. Application to the three-dimensional dual semigeostrophic equations. In this section, we apply Theorem 1.1 to the dual semigeostrophic equations on the three-dimensional torus $\mathbb{T}^3$. As introduced above, the dual formulation is

$$\left\{ \begin{array}{ll} \partial_t \rho_t(x) + \text{div}(\rho_t(x) U_t(x)) = 0, & (t, x) \in (0, \infty) \times \mathbb{T}^3, \\ U_t(x) = J(x - DP_t^*(x)), & (t, x) \in (0, \infty) \times \mathbb{T}^3, \\ \det D^2 P_t^*(x) = \rho_t(x), & (t, x) \in (0, \infty) \times \mathbb{T}^3, \\ P_t^* \text{ is convex,} & \\ \rho_0(x) = \rho^0(x), & x \in \mathbb{T}^3, \end{array} \right.$$

with the periodicity condition

$$(6.1) \quad P_t^*(x) - \frac{1}{2}|x|^2 \quad \text{is } \mathbb{Z}^3\text{-periodic.}$$

Here J is the matrix appearing in (1.7). Equivalently, $U_t = (x - DP_t^*)^\perp$, where $w^\perp = (-w_2, w_1, 0)$ for $w = (w_1, w_2, w_3) \in \mathbb{R}^3$. The initial density ρ^0 is a probability density on $\mathbb{T}^3$.

Global weak solutions for semigeostrophic systems in dual variables were constructed in [3, 9, 32]; see also [2] for the three-dimensional theory. When the initial density is Hölder continuous and uniformly bounded away from zero and infinity, Loeper [31] proved short-time existence and uniqueness in a more regular class. As in Subsection 1.2, our concern here is the time regularity under only the two-sided density bounds.

We now prove Theorem 1.6. The argument follows [22, Section 4] (see also [26, Theorems 15.12 and 15.13]). The only substantive change is that the three-dimensional equation for $\partial_t P_t^*$ is now covered by Theorem 1.1; the estimates used to control the dual flow are the three-dimensional analogues from [2].

Proof of Theorem 1.6. First, the density bounds are propagated by the dual flow. In particular, by [30, Theorem 9.1],

$$(6.2) \quad \lambda \leq \rho_t \leq \Lambda \quad \text{in } \mathbb{T}^3 \quad \text{for all } t \geq 0.$$

As in (1.10), differentiating the Monge–Ampère equation in time gives

$$(6.3) \quad \operatorname{div}(\rho_t(D^2 P_t^*)^{-1} D \partial_t P_t^*) = \operatorname{div} F_t, \quad F_t = -\rho_t U_t.$$

The periodic transport estimates (see, for example, [2, Theorem 3.1(iii)]) imply

$$(6.4) \quad \|F_t\|_{L^\infty(\mathbb{T}^3)} \leq C = C(\lambda, \Lambda) \quad \text{for all } t \geq 0.$$

Using the idea of Loeper [30, Theorem 5.1], together with the arguments in [1, Proposition 3.3] and [2, Proposition 1], there exist $\kappa = \kappa(\lambda, \Lambda) > 0$ and $C = C(\lambda, \Lambda)$ such that

$$(6.5) \quad \int_{\mathbb{T}^3} |D \partial_t P_t^*|^{1+\kappa} dx \leq C \quad \text{for all } t \geq 0.$$

One can also obtain (6.2)–(6.5) from [26, Theorem 15.12].

We fix the time-dependent additive constant in P_t^* so that

$$\int_{\mathbb{T}^3} \partial_t P_t^* dx = 0 \quad \text{for all } t \geq 0.$$

Poincaré’s inequality and (6.5) then give

$$(6.6) \quad \|\partial_t P_t^*\|_{L^{1+\kappa}(\mathbb{T}^3)} \leq C \quad \text{for all } t \geq 0.$$

We lift (6.3) periodically to $\mathbb{R}^3$. The periodic normalization (6.1) implies a uniform bound for $DP_t^*(x) - x$ on a fundamental cell, and hence fixed-height sections of P_t^* have uniformly controlled diameter. Therefore the structural constants in Theorem 1.1 can be chosen uniformly in t and in the center of the section. Applying that theorem with $p = 1 + \kappa$, using (6.2), (6.4), and (6.6), and covering a fundamental cell by finitely many interior sections, we obtain some $\gamma = \gamma(\lambda, \Lambda) \in (0, 1)$ such that

$$(6.7) \quad \|\partial_t P_t^*\|_{L^\infty((0, \infty); C^\gamma(\mathbb{T}^3))} \leq C.$$

Finally, let P_t be the Legendre transform of P_t^* . Differentiating the Legendre identity at corresponding points gives

$$(6.8) \quad \partial_t P_t(x) = -\partial_t P_t^*(DP_t(x)).$$

The two-sided bounds (6.2) and the periodic Monge–Ampère regularity theory give a uniform $C^{1,\beta}$ estimate for P_t , for some $\beta = \beta(\lambda, \Lambda) \in (0, 1)$. Combining this with (6.7) and (6.8), we obtain

$$|\partial_t P_t(x) - \partial_t P_t(y)| \leq C |DP_t(x) - DP_t(y)|^\gamma \leq C |x - y|^{\beta\gamma} \quad \text{for all } t \geq 0.$$

After decreasing the exponent if necessary, this proves (1.11). $\square$

6.2. Negative-norm stability and periodic homogenization. In this section, we prove Theorems 1.8 and 1.7.

Throughout this section, $n \geq 2$ and $\Omega \subset \mathbb{R}^n$ is a fixed bounded smooth uniformly convex domain. Fix also $\varphi \in C^3(\partial\Omega)$. For $f \in L^\infty(\Omega)$ with $\lambda \leq f \leq \Lambda$, let u_f be the convex Aleksandrov solution of

$$\det D^2 u_f = f \quad \text{in } \Omega, \quad u_f = \varphi \quad \text{on } \partial\Omega.$$

Existence, uniqueness, and stability of these solutions are recalled in [26].

Proof of Theorem 1.8. Fix $\mathbf{F} \in L^\infty(\Omega; \mathbb{R}^n)$ with $\operatorname{div} \mathbf{F} = f_1 - f_0$.

We show that there exist an exponent $\gamma \in (0, 1)$ and a constant $C > 0$, depending only on $n, \lambda, \Lambda, \Omega$, and $\|\varphi\|_{C^3(\partial\Omega)}$, such that

$$(6.9) \quad \|u_1 - u_0\|_{C^\gamma(\bar{\Omega})} \leq C \|\mathbf{F}\|_{L^\infty(\Omega)}.$$

Step 1: Smooth densities. First assume that f_0, f_1 are smooth on $\bar{\Omega}$.

By the global $C^{2,\sigma}(\bar{\Omega})$ solvability of the Monge–Ampère equation [40], the problems

$$(6.10) \quad \det D^2 u_t = (1-t)f_0 + tf_1 \quad \text{in } \Omega, \quad u_t = \varphi \quad \text{on } \partial\Omega, \quad 0 \leq t \leq 1,$$

admit classical uniformly convex solutions $u \in C^{2,\sigma}(\bar{\Omega})$ for all $0 < \sigma < 1$. Note that the density stays between λ and Λ and the boundary datum is the fixed function φ .

For each fixed t , the linearization of the Monge–Ampère operator at u_t is an isomorphism from $C_0^{2,\sigma}(\bar{\Omega})$ to $C^\sigma(\bar{\Omega})$, where $0 < \sigma < 1$; see [15, Chapter 6]. The implicit-function theorem and uniqueness therefore show that $t \mapsto u_t$ is C^1 . Differentiating (6.10), we obtain

$$L_{u_t} v_t = \operatorname{div} \mathbf{F} \quad \text{in } \Omega, \quad v_t = 0 \quad \text{on } \partial\Omega \quad \text{where } v_t = \partial_t u_t.$$

For $n \geq 3$, take $q = q_b + 1$. Since $\mathbf{F} \in L^\infty(\Omega) \subset L^q(\Omega)$, Theorem 1.4, applied with zero boundary datum, gives an exponent $\gamma > 0$ and a constant $C > 0$, both uniform in t and depending only on $n, \lambda, \Lambda, \Omega$, and $\|\varphi\|_{C^3(\partial\Omega)}$, such that

$$(6.11) \quad \|v_t\|_{C^\gamma(\bar{\Omega})} \leq C \|\mathbf{F}\|_{L^q(\Omega)} \leq C \|\mathbf{F}\|_{L^\infty(\Omega)}, \quad 0 \leq t \leq 1.$$

In dimension two, the same estimate follows from [23, Theorem 1.2].

Since

$$u_1 - u_0 = \int_0^1 v_t \, dt,$$

integrating (6.11) (or its two-dimensional analogue) establishes (6.9) for smooth densities.

Step 2: Bounded measurable densities. We use an approximation that preserves the divergence identity on the fixed domain. After a translation, we can assume that $B_r(0) \subset \Omega$ for some

$r > 0$. For $0 < \delta < 1/2$, let $a_\delta = 1 - \delta$ and $s_\delta = \delta r/4$. The convexity of Ω then gives $a_\delta \bar{\Omega} + B_{\delta r}(0) \subset \Omega$. For a nonnegative standard mollifier η_{s_δ} , define

$$f_{i,\delta}(x) = (f_i * \eta_{s_\delta})(a_\delta x), \quad \mathbf{F}_\delta(x) = a_\delta^{-1}(\mathbf{F} * \eta_{s_\delta})(a_\delta x).$$

The convolutions only sample points of Ω . Hence the approximations are smooth near $\bar{\Omega}$ and satisfy

$$\lambda \leq f_{i,\delta} \leq \Lambda, \quad \operatorname{div} \mathbf{F}_\delta = f_{1,\delta} - f_{0,\delta}, \quad \|\mathbf{F}_\delta\|_\infty \leq a_\delta^{-1} \|\mathbf{F}\|_\infty.$$

Moreover, $f_{i,\delta} \rightarrow f_i$ in $L^1(\Omega)$ by continuity of mollification and dilation in L^1 .

Let $u_{i,\delta} \in C(\bar{\Omega})$ be the convex solution to

$$\det D^2 u_{i,\delta} = f_{i,\delta} \quad \text{in } \Omega, \quad u_{i,\delta} = \varphi \quad \text{on } \partial\Omega.$$

From the stability of Aleksandrov solutions with a fixed boundary data, the weak continuity of Monge–Ampère measures, and the uniqueness of the Monge–Ampère equation, we find that $u_{i,\delta} \rightarrow u_i$ uniformly on $\bar{\Omega}$; see [26, Chapter 3]. Applying (6.9) to $u_{i,\delta}$ and $\mathbf{F}_\delta$, we have

$$\|u_{1,\delta} - u_{0,\delta}\|_{C^\gamma(\bar{\Omega})} \leq C \|\mathbf{F}_\delta\|_{L^\infty(\Omega)}.$$

Letting $\delta \rightarrow 0^+$, and using the uniform convergence and lower semicontinuity of the Hölder norm, we obtain (6.9).

Finally, taking the infimum over $\mathbf{F}$ in (6.9) completes the proof of (1.16). The theorem is proved. $\square$

We are now ready to establish a linear convergence rate in periodic homogenization for the Monge–Ampère equation.

Proof of Theorem 1.7. Since $f - \bar{f}$ belongs to $L^\infty(\mathbb{T}^n)$ and has mean zero, the problem

$$\Delta_y \psi = f - \bar{f} \quad \text{on } \mathbb{T}^n, \quad \int_{\mathbb{T}^n} \psi \, dx = 0$$

has a unique solution $\psi \in W^{2,p}(\mathbb{T}^n)$ for all $p \in (1, \infty)$ with appropriate estimates; see [15, Theorem 9.15 and Lemma 9.17]. In particular, for $p = 2n$, we have

$$\|\psi\|_{W^{2,2n}(\mathbb{T}^n)} \leq C(n) \|f - \bar{f}\|_{L^\infty(\mathbb{T}^n)}.$$

Let

$$G := D_y \psi.$$

Then, the Sobolev embedding (see [15, Section 7.7]) gives

$$\operatorname{div}_y G = f - \bar{f} \quad \text{on } \mathbb{T}^n, \quad \|G\|_{L^\infty(\mathbb{T}^n)} \leq C(n) \|f - \bar{f}\|_{L^\infty(\mathbb{T}^n)}.$$

Set

$$\mathbf{F}_\varepsilon(x) := \varepsilon G(x/\varepsilon).$$

Then

$$\operatorname{div} \mathbf{F}_\varepsilon = f(x/\varepsilon) - \bar{f}, \quad \|\mathbf{F}_\varepsilon\|_{L^\infty(\Omega)} \leq C(n) \varepsilon \|f - \bar{f}\|_{L^\infty(\mathbb{T}^n)}.$$

Clearly,

$$\|f(\cdot/\varepsilon) - \bar{f}\|_{\operatorname{div}, \infty; \Omega} \leq \|\mathbf{F}_\varepsilon\|_{L^\infty(\Omega)} \leq C(n) \varepsilon \|f - \bar{f}\|_{L^\infty(\mathbb{T}^n)}.$$

Since $\lambda \leq f, \bar{f} \leq \Lambda$ and both u_ε and $\bar{u}$ have the same boundary value φ , Theorem 1.8 proves both (1.14) and (1.15). The theorem is proved. $\square$

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